

The Impact of Environmental Taxes on Commercial Traffic and Its Environmental Consequences^{*}

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Abstract: This paper examines how commercial trucking responds to cross-country differences in environmental taxes on diesel fuel and whether these responses undermine the taxes' environmental objectives. We combine administrative data on truck traffic at German borders with data on individual freight journeys across Europe. We find that higher diesel taxes reduce truck traffic through the affected country. However, route-level evidence shows that trucks also respond by deviating from the fastest and shortest available routes to avoid countries with higher taxes and travel through countries with lower taxes. These detours increase the distance traveled. We further document the environmental consequences of this route substitution. Tax-induced increases in cross-border truck traffic raise local particulate pollution near affected border roads, while the additional distance traveled generates up to 1 million metric tons of CO₂ emissions. The findings show that when diesel tax rates differ across neighboring countries, commercial carriers may reduce their tax burden by reallocating traffic rather than reducing fuel consumption. Consequently, uncoordinated environmental taxes may shift traffic and pollution across borders while generating additional emissions through longer routes.

Keywords: Environmental Taxes, Truck Traffic, Mobility, Tax Avoidance, Emissions

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1 Introduction

The need to address the global climate crisis has sparked considerable interest in exploring effective policy instruments that can mitigate its consequences (IPCC, [2021](#)). As the transportation sector accounts for a significant portion of greenhouse gas emissions world-wide (European Environment Agency, [2022](#); International Energy Agency, [2022](#)), finding strategies to reduce emissions from commercial trucks has become a critical objective for policymakers and researchers alike (see e.g., World Bank, [2014](#); OECD, [2021](#); European Commission, [2023b](#)). Environmental taxes on diesel fuel have emerged as a potential instrument to incentivize more sustainable transportation practices and discourage high-emission activities (International Energy Agency, [2022](#)).¹ However, understanding how commercial traffic responds to environmental taxes, particularly when they vary across national borders, is a complex and so far understudied area. In this study, we aim to fill this gap.

The transportation sector is a significant contributor to global greenhouse gas emissions, accounting for 20% of the total emissions. Within this sector, road transport is responsible for the majority of this burden, accounting for three-quarters of emissions, which equals to 15% of the overall global emissions (International Energy Agency, [2022](#)). Trucks, in particular, heavily rely on diesel fuel, notable for its substantially higher carbon dioxide (CO₂) emissions compared to the majority of passenger cars fueled by gasoline. Fuel costs, constituting roughly 20% of a truck's total operating expenses, play a crucial role in companies' strategic route planning. Taxes, which account for nearly half of the gross fuel price, wield a decisive influence over where transport companies choose to refuel. This influence becomes particularly pronounced when tax rates vary across countries, as observed in the EU (Transport and Environment, [2021](#)), or across states, such as in the U.S., incentivizing commercial trucks to strategically plan their refueling locations. Environmental taxes on

¹ An alternative instrument is the implementation of a cap-and-trade system. So far, no cap-and-trade system has been implemented in the road transport sector.

fuel are designed to curb its demand, thereby aiming to reduce emissions and encourage a transition to cleaner alternatives. However, the existence of tax differentials across borders can lead to unintended consequences, such as trucks taking detours to access cheaper fuel in neighboring countries.

In this paper, we investigate how commercial trucks respond to differences in environmental taxes on diesel fuel across European borders. Our analysis proceeds in three steps to comprehensively assess the behavioral responses of truck operators to changes in tax rates and the resulting environmental consequences of such tax-induced responses. First, we use administrative toll data to examine whether cross-border truck traffic responds to changes in environmental taxes across countries. Next, we examine whether differences in tax rates between countries induce trucks to take detours to avoid high environmental taxes on fuel. Finally, we examine the environmental consequences of these actions by incorporating air pollution data and emissions.

We start our analysis by examining cross-border truck flows at the German borders. We obtain comprehensive administrative data containing monthly truck counts from mid-2018 to the end of 2022 on detailed road segments in Germany, which allows us to measure the responsiveness of cross-border truck flows between Germany and its neighboring countries to differences in environmental taxes on fuel. To overcome the challenge of isolating the impact of environmental taxes on the gross price of diesel, we use an instrumental variable (IV) approach. We leverage the environmental tax differential between Germany's neighboring countries and Germany as an instrument to predict tax-induced changes in the gross price differential of diesel. These predicted price changes resulting from changes in environmental taxes on diesel allow us to estimate the tax-induced effects on cross-border transport.

We find that commercial traffic responds to changes in environmental taxes. An increase in the tax rate of a neighboring country leads to a decrease in truck traffic through that country. Specifically, we observe that a tax-induced increase in the average gross

price of diesel fuel by one euro cent per liter of fuel results in 466 fewer trucks per road segment at the border per month, suggesting a shift in truck traffic patterns away from more expensive cross-border routes. This finding implies that trucking companies prioritize fuel cost-optimized routes and underscores the influence of environmental taxes on their decision-making processes. Moreover, we show that this effect is more pronounced on highways that are heavily used by commercial traffic. The effect is consistently observed within the first six road-sections close to the border, but gradually diminishes as one moves away from the border region and more inland traffic intersects, underscoring the robustness of our results.

Subsequently, we focus on individual truck routes across Europe. We leverage an administrative survey dataset spanning from 2011 to 2020 to investigate whether trucks strategically choose detours to avoid countries with high environmental taxes while taking advantage of low environmental taxes in other countries. The European Road Freight Transport (ERFT) database allows us to track individual truck journeys from their starting point to their end point, while recording the countries crossed in between. To determine whether trucks intentionally detour to avoid high fuel taxes, we compare the actual route observed in our data with the fastest and shortest route obtained from Google Maps (GMAP) via web scraping. Our survey dataset then includes the actual distance traveled per trip, the optimal distance, and the deviation between the two.

Leveraging environmental taxes on diesel fuel, averaged for each journey, we incorporate a linear probability model into our IV approach to examine whether an increase in the average environmental tax rate on diesel fuel for a journey affects the probability that a truck deviates from the fastest and shortest route available. Our results indicate that an increase in the average environmental tax on diesel significantly increases the probability that a truck deviates from the optimal route for the trip, suggesting that commercial trucking companies are indeed driving detours to save on fuel costs. Furthermore, we find that this behavior leads to an increase in total mileage, which includes both excess kilometers

and excess ton-kilometers, taking into account the size and environmental impact of the truck involved in the trip.

To further validate these findings, we take a closer look at the countries where detours are most common. Descriptively, we find that Luxembourg, along with Romania and Slovakia, are the main destinations for truck detours. Notably, these countries consistently have the lowest environmental tax rates on diesel on average in our dataset. To provide empirical support for this observation, we focus on a subset of cross-border trips that involve detours. We regress the number of divergent truck journeys per country and quarter on the average environmental tax rate for that country and quarter. We include country and year-quarter fixed effects to account for the size of a country and general time trends. Our results show a negative relationship, meaning that the lower the environmental tax rate on diesel, the more trucks pass through a country. This supports the notion that environmental taxes on diesel have a significant impact on commercial trucking by influencing route planning decisions to minimize diesel costs.

Finally, we assess the environmental consequences of tax-induced responses in the context of cross-border truck traffic. Expanding on our initial study of truck flows at German borders, we incorporate sensor stations that measure local air pollution to investigate whether increased cross-border truck flows resulting from environmental tax differentials contribute to local increases in air pollution. The sensor data we collect provide information on short-lived climate pollutants (SLCPs), with a particular focus on particulate matter (PM) emissions. PM is an important pollutant emitted during the combustion of fossil fuels that has adverse effects on both the environment and human health. Using an event study approach, we find that tax-induced increases in cross-border truck flows lead to local emission peaks at these border road-sections.

In addition, we use the results of our second analysis to quantify the environmental externalities resulting from the deviation of trucks to minimize fuel taxes. To do so, we establish a correlation between the excess ton-kilometers driven by trucks and the resulting

emissions. Building on our earlier results, we employ a linear prediction model to estimate the cumulative excess ton-kilometers resulting from tax-induced deviations. We then calculate the associated CO₂ emissions by multiplying the total surplus by its emission factor. We find an increase in emissions of up to 1 million tons of CO₂ due to trucks taking detours to avoid high environmental taxes on diesel fuel.

Our study provides evidence on the effects of environmental taxes on commercial transport and the resulting environmental impacts. Such evidence is scarce but policy-relevant as policymakers seek to promote sustainable practices in the transport sector.² We uncover unintended consequences arising from unharmonized tax rates within the EU, particularly in a sector that is crucial to the EU's climate change (European Commission, 2020). While our initial findings suggest that higher environmental taxes on diesel fuel lead to a reduction in commercial transport, in line with the objective of environmental taxes to reduce emissions from transport, we find in a subsequent analysis that trucks resort to deviations to avoid these higher taxes. As a result, they refuel in countries with lower tax rates, leading to increased mileage, fuel burned and thus higher overall emissions.

We contribute to the literature studying behavioral responses to environmental taxes in the transportation sector. The existing literature is relatively limited, with a notable absence of studies focusing on commercial transportation. Most research on environmental taxes in the transport sector has focused on non-commercial drivers, examining aspects such as fuel consumption and consumer responses to tax and price changes (Rietveld et al., 2001; Banfi et al., 2005; Davis and Kilian, 2011; Rivers and Schaufele, 2015; Jansen and Jonker, 2018; Bonnet et al., 2024). Notably, studies like Marion and Muehlegger (2011) and Li et al. (2014) utilizing fuel consumption data, find that consumers tend to react more strongly to tax changes than to price changes. Gimenez-Nadal and Molina (2019) use

² Alternative fuel options for the road transport sector will not be available in the foreseeable future, primarily because freight transportation typically involves heavy loads, making electricity, for example, impractical. Additionally, shifting transportation to railways is not feasible due to the lack of sufficient freight rail infrastructure in most European countries.

variation in gasoline tax rates across US (United States) states and finds that higher taxes lead to a greater number of people choosing public transportation instead of private cars for their commutes. Andersson (2019) examines the effect of a carbon tax on emissions in the Swedish transport sector and finds a reduction in pollution. Our study fills a critical gap in research by addressing how commercial truck traffic reacts to cross-border changes in environmental taxes across borders.³ We examine the mechanism that primarily affects fuel demand, namely driving behavior, using comprehensive data on traffic across Europe. This approach further distinguishes our study from previous literature that finds a net reduction in emissions in response to environmental taxes on fuel (e.g., Andersson, 2019; Runst and Höhle, 2022), thereby overlooking shifts in driving behavior and emissions when only examining demand and emissions at the implementation location of the tax.⁴

Prior literature examining firms' responses to environmental taxes in different settings finds that firms generally reduce their own emissions in response to taxes and cap-and-trade systems (Martin et al., 2014; Dussaux, 2020; Erbertseder et al., 2023; Martinsen et al., 2024). However, consistent with our study, firms reallocate resources to avoid paying or bearing environmental taxes and regulations without necessarily reducing emissions (Itzhak et al., 2021; Dechezleprêtre et al., 2022; Ecker and Keeve, 2023; Kaenzig et al., 2023; Colmer et al., 2024). We add to this literature by documenting evidence of this relationship for commercial trucking companies. Using extensive but unexplored traffic data combined with emissions data, we quantify the impact on truck traffic and the excess kilometers driven by trucking companies to avoid high environmental taxes on fuel. We further estimate the incremental emissions caused by this shift in driving behavior.

Furthermore, our study adds to the broader literature on the effects of taxes on mobility, drawing insights from works by Kleven et al. (2013), Schmidheiny and Slotwinski

³ A related study by Luechinger and Roth (2016) differs from our study in that it examines how trucking in Switzerland responds to the introduction of a mileage tax that differs from environmental taxes on diesel fuel.

⁴ Santos (2017) examines fuel taxes in Europe and calculates corrective fuel taxes aimed at internalizing externalities from road transport.

(2018), and Akcigit et al. (2016). This body of research has demonstrated that tax differentials and the mobility of the tax base can trigger various behavioral responses by individuals, including tax avoidance strategies. Our study builds upon this literature by showing that commercial truck drivers adjust their routes in response to cross-border tax differences, particularly within the context of environmental taxes.

2 Setting

2.1 Road Freight Transport in the EU

Road freight transportation plays a significant role within the transport sector, accounting for a substantial share of 76.3% of total freight transportation in the EU as of 2019 (European Commission, 2021). Emissions stemming from road transport make a considerable contribution, comprising approximately 15% of global CO₂ emissions (European Environment Agency, 2022; International Energy Agency, 2022). Trucks, in particular, heavily rely on diesel fuel, which is associated with higher levels of emittants compared to other fuels. Diesel engines emit a complex mixture of air pollutants, including CO₂, diesel particulate matter and nitrogen oxide (NO_x), which contribute to both adverse health effects and climate change (OECD, 2019).

A single truck tank filling can hold between 1,000 to 1,500 liters of diesel, allowing trucks to cover distances of up to 1,600 km.⁵ On average, a truck filling in Europe costs between EUR 1,400 and EUR 2,100. Thus, fuel costs represent a substantial portion, approximately 20%, of a truck's overall operating expenses, making it a critical factor in optimizing cost-efficient truck route planning (Trego and Murray, 2010; Hooper and Murray, 2018). Importantly, environmental taxes on fuel make a significant contribution to the total fuel price, accounting for up to half of it (see Appendix Figure A.1 and Figure

⁵ A distance of approximately 1,600 km can cover journeys such as traveling from Warsaw to Paris, from Amsterdam to Barcelona or from Hamburg to Rome.

A.2). As these taxes are a function of the gross fuel price, they exert a substantial influence on the refueling decisions made by transportation companies (Izadi et al., 2020).

Refueling given its substantial cost is typically not left to the discretion of individual truck drivers in commercial trucking businesses. Instead, it is planned and managed by the company itself. Truck transport companies aim to strategically minimize fuel expenses by employing commercial transportation management systems. These systems plan truck journeys on the go, considering factors like saving fuel costs along with truck driver working hours, the average speed of the truck as well as road congestions. Anecdotal evidence further supports this notion, with several business models advertising diesel price tools designed to plan fuel-optimal routes across Europe and highlight the lowest diesel prices available (see, e.g. [DKV Mobility](#)⁶ and [Transport Topics](#)).

Conversely, while road tolls are present in most EU countries, they are not among the cost factors that transport companies can easily avoid.⁷ These tolls apply to both highways and federal roads, making it difficult for trucking companies to consistently navigate around them. To operate profitably, transport companies must select routes on well-developed roads. Choosing alternative routes on less-developed roads typically incurs a time surcharge that is more expensive than the toll costs. Additionally, toll charges remained stable throughout the EU during our sample period.

2.2 Environmental Taxes on Fuel

Environmental taxes on fuel are at the core of taxation within the transportation sector (OECD, 2019). These taxes typically serve a dual purpose. Firstly, they are primarily designed to address the environmental externalities associated with fuel consumption. Secondly, they generate revenue for the government. Taxes on fuel typically encompass both value-added tax (VAT) and excise taxes reflecting the environmental impact associ-

⁶ Appendix Figure A.3.

⁷ Two different toll systems exist in the EU. Tolls are levied according to the distance travelled and the type of vehicle, while vignettes are scaled according to the duration of the use of the infrastructure.

ated with fuel usage. At the pump excise taxes are added to the net fuel price, with the resulting sum being multiplied by the applicable VAT rate.⁸ It is important to note here that VAT is not relevant in the context of freight transport, as companies can reclaim the VAT paid. As a result, we focus on excise taxes on fuel. In practice, different tax rates apply to the most commonly used fuels, diesel and gasoline. While diesel is predominantly used by trucks, gasoline is more common with private cars.

The landscape of environmental taxes on fuel (both diesel and gasoline) differs across nations. For instance, certain countries label their excise tax on fuel as mineral oil tax (e.g. Switzerland or Ireland), others include them in their energy tax (e.g. Germany). Either are applied per liter of fuel. Additionally, some countries impose a carbon tax based on the amount of CO₂ emissions produced during fuel combustion.⁹ We combine all these taxes as environmental taxes on fuel (European Commission, 2023a). The revenue generated from these environmental fuel taxes typically flows into the state's general budget, contributing to the financial resources of the government.¹⁰ In 2021, tax revenue from environmental taxes on fuel within the EU amounted to EUR 60 billion, representing 18% of overall tax revenue from environmental taxes (European Commission, 2023b).¹¹

As we narrow our focus to commercial traffic, our attention centers on environmental taxes concerning diesel fuel. Notably, environmental tax rates on diesel vary significantly between EU member states, with certain countries imposing lower taxes compared to others. These differences present opportunities for commercial truck companies to strategically plan their journeys, opting to refuel in countries with lower diesel tax rates, thereby minimizing their operational costs. In some EU countries, including Belgium, France, Hungary, Ireland, Italy, Romania, Slovenia, and Spain, commercial trucking businesses can

⁸ Gross price fuel = (Net price fuel + Environmental tax fuel) x (1 + VAT).

⁹ An illustration involves the introduction of a carbon tax on fuels in Luxembourg and Germany in 2021, followed by Austria in September 2022. Notably, Switzerland excludes fuel usage from its carbon tax, while Slovenia places the liability for the carbon tax on fuel producers rather than fuel consumers. Meanwhile, France and Finland embed an explicit carbon component within their fuel excise tax.

¹⁰ In contrast, the money generated by road tolls is used to maintain the road infrastructure.

¹¹ Other environmental taxes besides taxes on fuel are energy taxes, pollution taxes and resource taxes.

partially recover the diesel taxes they pay through state-issued tax rebates.¹²

We use these variations in environmental taxes and the availability of tax rebates to investigate the responsiveness of commercial trucks to differences in these tax regimes. Our analysis begins by examining cross-border traffic in Germany. Germany serves as an ideal setting due to its prominent role in intra-European land transport. Its geographical location at the heart of Europe coupled with a well-developed transportation infrastructure including highways and federal roads results in a constant flow of goods between Germany and its neighbors. Moreover, Germany's environmental tax on diesel remained stable throughout our sample period, making it an ideal choice as our baseline country.¹³ Figure 1 illustrates the monthly evolution of environmental taxes on diesel fuel in euro cent per liter across Germany and its neighboring countries from 2011 to 2022. Notably, Germany imposes rather high taxes on diesel, while Poland and Luxembourg consistently maintain lower rates. Additionally, Belgium and France have comparatively higher taxes on diesel fuel for cars but offer tax rebates, as depicted by the dashed line, for commercial trucks.¹⁴

Our first analysis leverages the variations in environmental taxes on diesel fuel relative to the German diesel tax rate. We subtract the environmental tax of Germany from that of each of its neighboring countries per month (tax in neighboring country - tax in Germany). Figure 1 presents the development of diesel taxes compared to Germany, encompassing Germany's neighboring countries and the time period mid-2018 to 2022 for which we possess traffic data from German roads. These relative differences in environmental tax rates on diesel serve as the main explanatory variable in the initial part of our empirical analysis, aiming to investigate the response of commercial traffic to changes in

¹² In 2018, the total amount of truck diesel tax rebates paid by these EU countries amounted to approximately EUR 4 billion. Currently, trucks in the EU pay an average diesel tax of EUR 0.43 per liter, which is EUR 0.07 below the rate imposed on cars (Transport and Environment, 2021).

¹³ Germany adjusted its environmental taxes on fuel only in 2022, a response prompted by the events surrounding the Ukraine war.

¹⁴ The drops observed in 2022 were implemented as a response to the Ukraine war, during which several nations reduced their taxes on fuel as a countermeasure against rising oil prices.

environmental taxes.

In our subsequent analysis, we delve into whether truck drivers take detours to circumvent high environmental taxes. For this purpose, we once again employ the diesel tax rates, now encompassing all continental European countries, and extending the time frame from 2011 to 2022. Table 1 shows summary statistics of (commercial) diesel tax rates by all European countries included in our sample. This expanded dataset on environmental tax rates on diesel allows us to examine truck routes throughout Europe and explore potential deviations undertaken by drivers to avoid regions with high environmental taxes.

3 Data and Descriptives

3.1 Traffic: Toll Data

To assess the impact of environmental tax rate differences on cross-border truck traffic flows between Germany and its neighboring countries, we utilize administrative toll data for Germany. The data provide us with monthly information on all trucks passing through Germany at a highly detailed spatial level, covering both highways and federal roads.¹⁵ By observing the monthly count of trucks traveling on each road-section per month, we can assess how sensitive truck traffic flow is to changes in environmental taxes on fuel. Specifically, we observe the number of trucks traversing a given road-section for each direction independently resulting in two monthly observations per road-section. The data available spans from mid-2018 to the end of 2022.¹⁶ Given that the data include information on the number of axes (i.e. truck size), we restrict our analyses to trucks with more than four axes, representing about 80% of the truck count in our sample (see Appendix Figure B.1). Road freight transportation heavily relies on trucks equipped with

¹⁵ In our study, we use the term highway to refer to the German "Autobahnen" while federal roads are German "Bundesstraßen".

¹⁶ Data availability is limited to mid 2018 until the end of 2022 as the German toll system for trucks was implemented incrementally. The implementation started in April 2017 on highways, and was extended to federal roads by July 2018. As a result, data for the period before July 2018 is not available for analysis.

four or more axes, which are also the predominant vehicles commonly observed on highways.¹⁷ Our data further includes the road type driven (i.e. highway vs. federal roads). This enables us to analyze heterogeneity in tax sensitivity between road types, with highways possibly being the preferred choice for truck drivers in terms of delivery speed and infrastructure.

Figure 2 displays all highways and federal roads in Germany included in our dataset, while Table 2 provides descriptive statistics such as the total number of observations, which amounts to 690,792, and the average length of a road-section, which stands at 731 meters. Notably, 96% of the road-sections in our dataset are composed of federal roads, while the remaining 4% pertain to highways. Road-sections on highways are, however, on average ten times longer than those on federal roads. Moreover, monthly truck count statistics underline the significance of German "Autobahnen" (highways), with a monthly average of 79,273 trucks passing through each highway section, in contrast to an average of 4,856 trucks on federal roads.

Given our focus on cross-border flows, Figure 2 (b) identifies the border corridors we use to examine the impact of environmental taxes on cross-border truck traffic. We analyze road-sections that directly cross the border, as well as those within 5 km, 10 km, and 15 km of the border. A total of 135 highways and federal roads intersect the German border, connecting to neighboring countries. These roads serve as crucial transportation links, facilitating cross-border movement and trade between Germany and its neighboring countries. Appendix Table B.1 presents the average monthly truck traffic flow by distance to the border, while also differentiating traffic intensity by country of destination.

¹⁷ Trucks featuring three axes, such as car cranes, and those with two axes, like tractors or combined harvesters, serve different specialized purposes.

3.2 Traffic: Survey Data

In our second analysis we utilize a large sample of administrative route survey data to track truck drivers on their journeys across Europe. The ERFT database provided by EURO-STAT enables us to trace individual truck drivers from their starting point to their final destination while capturing the countries they cross in between.¹⁸ This second dataset allows us to observe the actual routes commercial trucks drive through Europe. For anonymity purposes the starting and end point of each journey is reported in NUTS-2 (Nomenclature of Territorial Units for Statistics) format, which corresponds to district-sized geographical regions.¹⁹ Moreover, the ERFT database contains information on the actual distance traveled per journey (in kilometers), the type of journey (whether the truck is laden or unladen), and the type of transport (whether the journey was for hire or on own account). In total, we observe more than 46 million journeys from 426.115 unique vehicles during the time period from 2011 to 2020.²⁰ As we are interested in journeys that drive through continental Europe, we filter the data and find 1,182,339 cross-border journeys provided by 307,241 unique respondents.

Using this second dataset we investigate whether trucks deliberately take detours to avoid high environmental taxes. For this, we compare the actual route taken by each truck driver with the fastest and shortest route possible. The detour is quantified by measuring the difference between these two distances while also noting the specific countries trucks deviate to. Given that the ERFT dataset originally only provides information on the actual route taken for each journey, we employ a web scraping technique to extract the fastest and shortest route from GMAP for every journey. Our web scraper utilizes the journey's

¹⁸ According to a regulation by the European Commission, each member state is required to collect data on the carriage of goods by road. This data is obtained by surveying owners of road vehicles registered within their respective member states. It includes the journeys made by the vehicles, including the start and end points, as well as the countries crossed. Countries are required to report quarterly surveyed micro-data on selected vehicles. See Article 1 of Regulation (EC) 70/2012, as of 18 January 2012 (European Commission, 2012).

¹⁹ The NUTS format classifies Europe's territory in a hierarchical order, where NUTS-1 corresponds to larger regions such as federal states, while areas covered by NUTS-2 and NUTS-3 are more granular.

²⁰ Release date: 10 November 2021, version 1.

start and end points from the data, searching for the most direct route in GMAP. Since the start and end points are classified under the NUTS-2 system, we execute this search using the centroids of each NUTS-2 district. Additionally, we ensure that GMAP identifies a suitable location near these centroids, ensuring that simulated routes commence from areas suitable for road traffic. Simultaneously, we collect information on the countries traversed during each journey. This allows us to determine the potential detours taken for the specific journeys.

To validate the accuracy of our web scraping approach we show the properties of the actual ERFT data, including the actual distances driven, and the scraped GMAP data, including the fastest and shortest route feasible, in Figure 3. As Panel (a) depicts, we identify 1.4% of journeys as outliers, classifying them as such due to their significantly longer duration relative to the distance driven.²¹ We exclude these outliers from our sample, enhancing the reliability of our analysis (see Panel (b)). Figure 3 Panel (c) further illustrates a comparison of the distances of journeys between the ERFT and GMAP data. The alignment of these distances provides additional validation for our scraping approach. To address concerns regarding potential temporal biases in the GMAP data due to the time of scraping, Panel (d) displays the hourly averages of the scraped data. Notably, these averages remain consistent across different times of the day, mitigating any potential time-related influences on our results.

We present summary statistics for our refined survey dataset in Table 3, providing insights into both all cross-border journeys and those that involve deviations. On average, each journey covers a distance of 624.05 kilometers, while the optimal journey distance is 448.24 kilometers, resulting in an average deviation distance of 24.39 kilometers. Among the cross-border routes in our sample 71,793 deviate by traversing different or additional countries compared to the optimal route. The majority of journeys involve laden and hired trucks. The countries crossed most frequently are Germany, the Netherlands, and Belgium

²¹ These deviations are typically triggered by extraordinary events such as extensive traffic congestion, road closures, or truck breakdowns.

(see Appendix Table [C.3](#)).

3.3 Emissions: Sensor Data

In the second part of the paper we aim to estimate the environmental consequences of trucking businesses' tax-induced reactions. To link the tax-induced increases in truck traffic to local increases in environmental emissions, we rely on two data sources: first, we use administrative air pollution data provided by the German Federal Environment Agency (henceforth, we use the abbreviation of its German name "Umweltbundesamt", i.e. UBA). Second, we employ publicly available, open source (OS) air pollution data provided by the Sensor Community.²² Both databases collect sensor data on short-lived climate pollutants (SLCPs), with particulate matter (PM) being the key pollutant collected. PM is emitted when fossil fuel is combusted. Road traffic is the main source of PM (World Health Organization, [2005](#)). PM is classified in different categories of fineness (i.e. PM_{2.5}, PM₁₀) and is measured in particles (in g/m³). In our analyses we focus on PM₁₀ as it is the most consistently measured air pollutant in both datasets and combined with our traffic data. SLCPs such as PM have serious consequences for health and climate change. Black carbon, a component of fine particulate matter, is one of the largest contributors to global warming. It warms the earth's atmosphere by absorbing sunlight, thereby accelerating the melting of snow and ice (Ramanathan and Carmichael, [2008](#)). Moreover, PM can cause harm to human health. PM can be carried deep into the lungs where it can cause inflammation and worsen the condition of people with heart and lung diseases (European Environment Agency, [2023](#)).

To investigate the impact of increased truck traffic on environmental emissions, we combine our traffic data and emissions data. Leveraging our toll dataset, we link sensors

²² The [Sensor Community](#) project originated from [Luftdaten.info](#), an open knowledge initiative located in Stuttgart, Germany. It began as a regional project in 2015 but has grown significantly in recent years and now covers air pollution sensors across the world. Citizens install sensors that measure air pollutants, and the Sensor Community generates a constantly updating particulate matter map from the transmitted data.

to corresponding road-sections based on geographical coordinates. We hereby focus on road-sections located closer to the border (i.e. up to a road-order of 60 inside Germany). The advantage of the UBA data lies in its complete time coverage and measurement reliability, implying no outlier observations in any direction and a fully balanced panel. Conversely, the measurement data provided by OS is significantly larger, allowing us to identify emission responses to taxation at a very granular, local level. However, station coverage changes over time and the share of outliers for PM10 is substantial. We thus restrict our initial sample as follows: first, to reduce excessive imbalance over time, we require each OS emission station to have at least 40 year-month observations (equal to the sample median). Second, we winsorize the year-month values of PM10 at the level of both lower and upper 5%. This yields a similar tail distribution of PM10 as the UBA data. In order to balance the advantages and limitations of both emission data sources, for our further analysis we require that PM10 is measured at the closest UBA or OS emission sensor station.

Table 4 shows the summary statistics for the distance of a road-section to a sensor station and the monthly PM10 emissions for highways and federal roads. The median distance between a sensor and the nearest cross-border road-section is equal to 500 meters for the full sample, but increases to 6.4 kilometers when considering only cross-border road-sections. Given that PM can be reliably monitored within a range of up to 1,000 meters, this setup still allows us to evaluate environmental emissions for subsets of emission stations located closely to road-sections at highways and federal roads in Germany.

4 Effect of Environmental Taxes on Commercial Traffic

4.1 Toll Data

Empirical Strategy. To investigate the effect of environmental taxes on cross-border truck traffic, we examine changes in the environmental tax differential between Germany and its neighboring countries. Environmental taxes on diesel fuel influence a trucking

company's refueling choices by being part of the gross price of diesel. Diesel prices are endogenously determined by market dynamics, which involve both supply and demand factors. Consequently, the gross price of diesel is influenced not only by the environmental tax rate but also by variables such as oil prices and decisions made by local fuel station owners. To address endogeneity issues due to simultaneity or imperfect pass-through and to isolate the variation in the diesel gross price attributable to environmental taxes, we employ an instrumental variable strategy. We instrument the gross price differential with the environmental tax differential (see a similar approach e.g. in, Bonnet et al., [2024](#)). This enables us to isolate the tax-induced impact of gross diesel price differences on cross-border truck traffic.

An effective instrument must satisfy two key criteria. First, the instrument must be correlated with the endogenous variable (*relevance condition*). In our context, this means that environmental taxes on diesel must be correlated with the gross price of diesel. Our first stage regression (see Table [5](#) Panel A) confirms this correlation. Second, the *exclusion restriction* necessitates that, conditional on covariates, the instrument influences the outcome only through the endogenous variable. In our case, environmental taxes on diesel exclusively affect cross-border truck traffic through their influence on the gross price of diesel. Given that environmental taxes are integrated into the gross price and that truck drivers ultimately consider the gross price when making decisions, this assumption appears reasonable. Moreover, as environmental tax revenues are not used to maintain road infrastructure, we can mitigate concerns that road quality is a correlated omitted variable.

One remaining identification challenge is the potential policy endogeneity and omitted variable bias that arise from using policies as sources of variation. To address this, we provide evidence that our tax variable is plausibly exogenous. Figure [A.4](#) shows that the average monthly net and gross price of diesel in Germany and its neighboring countries follows a similar pattern during our sample period. This suggests that the net price of

diesel is primarily influenced by the oil price, while gross price differences between countries are driven by changes in tax rates. We further conduct an additional robustness check excluding the months following the outbreak of the Ukraine war (February 2022). In response to the war, many countries significantly lowered their taxes on diesel, potentially leading to confounding. Our results remain consistent (see Table B.4).

We implement the IV estimator using two-stage least squares (2SLS). In the first stage, we regress the gross price differential of diesel on our instrument, the environmental tax differential. The gross price and tax differentials are computed as the difference between the gross price or environmental tax in the neighboring country and the corresponding values in Germany. Germany is selected as our baseline country as we are examining traffic at German borders. Moreover its location, infrastructure, and relatively stable environmental tax on diesel fuel make it a suitable reference point. In the second stage, we regress our outcome variable of interest, commercial cross-border traffic, on the predicted gross price differential.

To estimate the first stage regression, we set up the following equation:

$$\text{Gross price differential}_{icm} = \alpha_i + \beta_m + \gamma \times \text{Env. tax differential}_{icm} + \varepsilon_{icm}. \quad (1)$$

$\text{Gross price differential}_{icm}$ represents the diesel gross price difference between country c and Germany on road-section i during a particular year and month m . Our instrument $\text{Env. tax differential}_{icm}$ captures the variation in environmental taxes on diesel. Using the predicted values of the first stage regression, we estimate the following second stage equation:

$$\text{Cross-border traffic}_{icm} = \alpha_i + \beta_m + \delta \times \widehat{\text{Gross price differential}}_{icm} + \varepsilon_{icm}. \quad (2)$$

$\text{Cross-border traffic}_{icm}$ represents the number of trucks driving on a specific cross-border road-section i during a particular year and month m .²³ Our variable of interest is the instrumented diesel tax rate differential $\widehat{\text{Gross price differential}}_{icm}$ that captures the tax-

²³ We refrain from using a Poisson specification because the second stage of our 2SLS analysis cannot be non-linear, even though our outcome variable is a count variable.

induced variation in diesel gross price between a neighboring country and Germany. By construction Equation 2 estimates the Local Average Treatment Effect (LATE). The coefficient of interest δ examines how a tax-induced change in the gross price differential of diesel affects commercial traffic flow at the border. Put differently, it estimates the effect on cross-border truck traffic for price changes that are caused by environmental tax changes on diesel. In all our specifications, we include road-section fixed effects (α_i) to account for factors that remain constant over time and are specific to a road-section. We also include year-month fixed effects (β_m) to control for variations in time and cluster standard errors by road-section.

Finally, to examine the direct effect of the environmental tax differential on cross-border traffic, we specify the following reduced form equation:

$$\text{Cross-border traffic}_{icm} = \alpha_i + \beta_m + \lambda \times \text{Env. tax differential}_{icm} + \varepsilon_{icm}. \quad (3)$$

Since truck traffic predominantly occurs on highways (see Table 2), we conduct our regression analyses separately for two groups. First, we examine the entire set of cross-border road-sections on highways and federal roads, which includes our full sample. Second, we narrow our focus to cross-border road-sections situated on highways, reflecting the most prominent roads for truck traffic. To broaden our perspective and provide a more comprehensive understanding of the effects, we further expand our analysis to include road-sections located at different distances from the border. To mitigate concerns that our results are confounded by the Covid-19 pandemic, we exclude affected months (March and April 2020) in a robustness test.²⁴

Results. Table 5 Panel A presents the results of the first stage regression, exploring the relation between the environmental tax differential on diesel and the gross price differential.

²⁴ To control for pandemic factors that equally affected cross-border traffic between countries during the same period, we include year-month effects. Thus, in this additional robustness test, we specifically test for the possibility that, for example, border closures affecting only a subset of neighboring countries could affect our results.

Column (1) displays the results for the full sample, while Column (2) focuses on highways. In both specifications, we observe a significant positive effect of the tax differential on the gross price differential. This suggests that our instrument, the environmental tax differential, accurately predicts the gross price differential. The high F-statistics support this notion and suggests that our instrument is not weak.

In Table 5 Panel B, we explore the reduced form directly regressing cross-border truck traffic on the environmental tax differential on diesel. Again, Column (1) presents results for the full sample, while Column (2) narrows the focus to highway traffic. The estimated coefficients are statistically significant showing that commercial cross-border traffic reacts negatively to an increase in the tax differential. An increase in the tax differential by one euro cent per liter of fuel (referring to an environmental tax increase in the neighboring country) is associated with an average decrease of 47 trucks per month on a cross-border road-section in the full sample, with this reduction increasing to 327 trucks per month on highways.

Table 5 Panel C presents results for the OLS specification, regressing cross-border traffic on the gross price differential for the full sample and highways. For the full sample, we find a positive coefficient that is statistically insignificant. In contrast, for the highway sample, we observe a significantly negative coefficient. Under our preferred IV specification in Panel D, both 2SLS coefficients are negative and statistically significant. We find that an increase in the environmental tax rate on diesel in a neighboring country leads to a substantial reduction in cross-border traffic toward and away from that country. More precisely, a tax-induced increase in the price differential by one euro cent per liter of fuel (i.e., the neighboring country experiences an environmental tax-induced increase in diesel price by one euro cent relative to Germany) is associated with an average monthly decrease of 53 trucks on all cross-border roads and a significant decrease of 466 trucks on highways. These findings align with our initial expectations, demonstrating that truck traffic responds to changes in environmental tax rates on fuel. An increase in the environmental tax rate

on diesel in one country decreases truck traffic flow through that country, while a decrease in the tax rate leads to an increase in traffic flow. This result is stronger on highways, highlighting the importance of highways for trucking companies.

To calculate the overall increase in truck traffic caused by tax differences, we consider the mean environmental tax rate difference of -6.5 euro cent per liter of fuel between two neighboring countries and multiply it by our estimated coefficient. We predict approximately 1,177 additional trucks per month per cross-border road-section and 3,029 additional trucks per month on highway road-sections. For Germany as a whole, this amounts to an annual excess of approximately $1,177 \text{ trucks} \times 12 \text{ months} \times 2 \text{ incoming and outgoing road-section} \times 133 \text{ cross-border roads}$, which results in an increase of around 4 million truck commutes per year. This highlights the substantial impact that even small shifts in environmental tax rates on diesel in neighboring countries can exert on the volume of cross-border truck traffic in Germany.

Next, we broaden our perspective by including road-sections located further away from the border. Figure 4 illustrates the 2SLS estimates on the tax-induced price differential by the distance to the border. Figure 4 (a) estimates Equation 2 separately by road-section. Road-section 0 is the section directly located at the border, while road-section 1 is one section away from the border. Notably, the coefficients for the first six sections are all negative and statistically significant. However, the magnitude of the coefficient diminishes as we move farther from the border. We attribute this trend to an increasing presence of inland traffic on road-sections located at a greater distance to the border. As more domestic traffic intersects with these roads, the impact of tax differentials on cross-border truck traffic gradually diminishes. In Figure 4 (b) we separately estimate point estimates for road-sections directly at the border, 0 to 5 km, 5 to 10 km, and 10 to 15 km from the border. We confirm that the negative effect gradually decreases the farther we move away from the border and add it is most pronounced within the first 5 kilometers to the border.

We acknowledge that our results could be contaminated by the impact of concurring

events. Our results are unlikely to be biased by the COVID-19 pandemic and associated border closures in March and April 2020. While borders were closed for private travelers, they remained open for transport vehicles. Additionally, some countries relaxed Sunday driving bans for trucks, opened green lanes for faster truck passage at borders, and eased driving and rest requirements. Therefore, we believe that the pandemic did not have a significant impact on transport operations due to these special measures. If trucks were unable to respond to environmental tax differences due to border closures during this period, our results would be biased toward zero. To address this concern, we exclude the first two months of the pandemic (March and April 2020) from our sample. We document in Appendix Table B.3 that our estimated coefficients remain consistent.

Another potential confounding event is the outbreak of the war in Ukraine, which caused fuel prices to spike throughout 2022. At the same time, European governments responded by lowering environmental taxes on fuel across the board. To ensure that our results are not affected by this confounding event in February 2022, we test whether our results hold when excluding all months after January 2022. We expect to find even stronger results for this shortened sample period, as tax changes that occurred due to the Ukraine are more closely aligned across countries. Indeed, in Appendix Table B.4 we observe even stronger responses when we exclude the period of the Ukraine war.

4.2 Survey Data

Empirical Strategy. Next, we investigate whether differences in tax rates across countries induce trucking companies to take detours. To this end, we examine individual truck journeys across continental Europe. We begin by observing the actual distance traveled by a truck during its trip and the countries it crosses, as recorded in the survey data. By leveraging data scraped from GMAP, which includes the fastest and shortest routes along with the associated countries, we can compare these two routes to determine whether a truck deviates from the optimal route and assess if this deviation is driven by fuel tax

differences.

First, we assess whether the likelihood of deviation increases with higher environmental taxes on diesel along the journey. We begin by creating a binary variable that indicates whether a truck's journey deviates from the fastest and shortest route available. A journey is classified as deviating if it involves driving a longer distance and traveling through an additional or different country compared to the optimal route. Subsequently, we regress this deviation indicator on the average gross price of diesel along the journey. To account for potential endogeneity, we instrument the gross price with the average environmental tax on diesel to isolate the tax-induced component of the gross diesel price. We then apply a linear probability model²⁵ to estimate the following 2SLS regression:

$$Deviation_{jnq} = \alpha_n + \beta_q + \delta \times \widehat{Gross\ price\ average}_{jnq} + \varepsilon_{jnq} \quad (4)$$

The dependent variable $Deviation_{jnq}$ is coded as one if the truck journey j deviates from the optimal route in quarter q by passing through an additional or different country, and zero otherwise. We define the $Gross\ price\ average_{jnq}$ as the average gross price of diesel along journey j in quarter q , instrumented by the average environmental diesel tax in the first stage. We expect a positive coefficient ($\delta > 0$) for our coefficient of interest δ , supporting the hypothesis that trucks are more likely to deviate when the average tax rate along the route increases. Our model incorporates robust standard errors. Additionally, we introduce year-quarter fixed effects (β_q) to capture time-specific variations and NUTS2-start-end-point pair fixed effects (α_n) to account for factors specific to a journey from one region to another. This addresses concerns that our estimated effect might be influenced by factors other than taxes, such as geographic characteristics or tolls, that could determine the optimal route. With this demanding fixed-effect structure, we account for variation in tax rates over time while holding the specific journey in NUTS2 classification constant.

²⁵ We choose a linear probability model over a logit or probit specification to avoid the incidental parameter problem, which would pose challenges in including fixed effects (Greene, 2002). For robustness, we provide the results using a probit specification without fixed effects in Appendix Table C.2, where our results remain consistent.

However, recognizing that fixing our results on NUTS2-start-end-point pairs leads to a loss of variation, we also present the results without any fixed effects.

Next, we quantify the additional distance traveled by trucks due to variations in environmental taxes. To achieve this, we replace the deviation indicator variable with the journey's deviation distance, measured both in kilometers and ton-kilometers to account for the size and environmental impact of the truck involved in a journey. On the right-hand side of Equation 4 we introduce an interaction term, which incorporates the deviation indicator. This adjustment allows us to quantify how far truck companies deviate when faced with an increase in environmental taxes on diesel. As before, we employ 2SLS to isolate the tax-induced changes in the gross price of diesel.

Subsequently, we seek to gain a more profound understanding of the countries through which trucks deviate from their optimal routes. Initially, we conduct a descriptive analysis to identify countries where detours are most prevalent. In this step, we compile our data into a country-quarter panel, capturing the number of journeys passing through a country and the number of deviations within that country. By computing the share of these two metrics, we gain insights into the countries most frequently deviated through by controlling for the overall number of trucks passing that country. We empirically assess whether these deviations are influenced by tax considerations. We regress the number of deviating truck journeys per country and quarter on the tax-induced average gross price of diesel for that specific country and quarter using our 2SLS model. Our specification incorporates county and year-quarter FEs accounting for variations in a country's size and across time.

Results. We initiate the analysis of the survey data using a linear probability model to assess whether differences in the environmental taxes on diesel influence the likelihood of truck companies deviating from the fastest and shortest route. The results are detailed in Table 6. Panel A displays the outcomes of the first stage, regressing the average diesel gross price on the average environmental tax on diesel. Panel B presents the results of

the reduced form, regressing the deviation indicator on the average environmental tax on diesel. We perform an OLS specification, directly regressing the deviation indicator on the gross price average of diesel, in Panel C. Finally, we present the second stage results, where the deviation indicator is regressed on the tax-induced average gross price of diesel, in Panel D. Columns (1) do not include fixed effects, and Columns (2) incorporate both year-quarter and NUTS2-start-end-point fixed effects.

To validate the relevance condition, Table 6 Panel A displays the results of the first stage regression. The positive and significant coefficients confirm the existence of a correlation and, consequently, thereby affirming the relevance of the instrument. Additionally, the sufficiently high F-statistics support this notion, confirming that our instrument is not weakly identified.

The results from the reduced form in Table 6 Panel B reveal a positive and significant relation between the probability of deviating from the shortest and fastest feasible route and the average environmental tax rate on diesel. This suggests that truck drivers are more inclined to deviate from the optimal journey when faced with higher average tax rates on that route. To be precise, a one euro cent per liter of fuel increase in the average environmental tax corresponds to a 0.43% (0.18%) increase in the probability of a truck deviating from its optimal route, without (with) the inclusion of fixed effects.

Finally, the results of the second stage regression in Table 6 Panel D validate the positive relationship highlighted by the reduced form. We observe that a tax-induced increase in the average gross diesel price by one euro cent per liter of fuel corresponds to a 0.97% (0.13%) increase in the probability of a detour, without (with) the inclusion of fixed effects. Initially, in our toll data analysis, we established that truck drivers respond to changes in environmental taxes by reducing traffic when taxes are raised. This second part demonstrates that our first results, a reduction in traffic causing less cross-border traffic, doesn't mean trucks are driving less overall. Instead, the probability of transport companies deviating to refuel in a country with lower environmental taxes on fuel increases

when differences in environmental taxes exist.

We delve deeper into the implications of these findings by quantifying the extent of the detour. To achieve this, we replace the deviation indicator variable with a variable containing the deviation distance and re-estimate our 2SLS IV regression, as illustrated in Table 7. Panel A shows the deviation distance in kilometers as the dependent variable, while Panel B measures deviation in ton-kilometers, accounting for the size and freight a truck is transporting. For the deviating journeys, we find that a tax-induced increase in the average gross diesel price by one euro cent per liter increases the deviation distance by 132 meters in the 2SLS specification and by 398 meters in the reduced form. Looking at excess ton-kilometers, we confirm a positive relationship, observing that a one euro cent per liter increase in the tax-induced average gross diesel price leads to an increase of 2.04 ton-kilometers.

Next, our focus shifts to identifying countries where detours are most prevalent. Initially, we descriptively examine the number of trucks deviating through a country, scaled by the overall number of trucks passing through that country. Our observations highlight Luxembourg, along with Romania and Slovakia, as primary destinations for truck detours (see Appendix Table C.3). These countries exhibit the lowest environmental tax rates on diesel on average in our dataset.

To empirically confirm this observation, we focus on a subset of cross-border journeys involving deviations. We regress the number of deviating truck journeys per country and quarter on the average tax-induced gross diesel price for that country and quarter. Our model incorporates controls for the total number of journeys passing through each country per quarter, as well as the country's population, to account for variations in size and traffic. The results in Table 8 reveal negative coefficient estimates, indicating that the volume of trucks deviating through a country increases as the environmental tax in that country decreases. This finding substantiates the notion that environmental taxes on diesel significantly impact commercial trucking operations, influencing route planning decisions

to minimize diesel costs.

5 Environmental Consequences

5.1 Combining Toll and Emission Data

To assess the environmental implications of tax-induced shifts in truck traffic, we link our toll dataset to emissions data collected by sensor stations near specific road-sections. Focusing on the German cross-border setting, we examine emissions from roads leaving and entering Germany. To gain a first understanding of the relationship between environmental taxes and emissions, we plot the tax differentials between Germany and its neighboring countries against the average emissions on cross-border roads over our study period (mid-2018 to 2022) in Figure 5. The figure suggests that PM10 emissions generally increase on average when the tax in the neighboring country decreases, particularly evident when the tax differential becomes more negative, favoring lower environmental taxes in neighboring countries (e.g., Austria, AT; Belgium, BE; Czech Republic, CZ; Denmark, DK; Netherlands, NL; Poland, PL). Conversely, we observe a decrease in average emissions during periods where the environmental tax differential increases, indicating that neighboring countries have raised their environmental taxes on fuel compared to Germany (e.g., Austria, AT; Belgium, BE; Switzerland, CH; Denmark, DK; Luxembourg, LU; Netherlands, NL; Poland, PL). We observe a shifting effect of emissions towards the country with lower tax rates.

To analyze how local emission pollution reacts to tax-induced variations in cross-border truck traffic, we again employ the reduced form approach already utilized in Equation 3, with PM10, measured at the closest emission sensor station, being the dependent variable. Our analysis is limited by the requirement that PM10 must be measured at a close distance, i.e. in a radius of not more than one kilometer. As Table 4 shows, however, this is only the case for a small share of our road-sections. We thus estimate our reduced

form empirical model for subsets of our overall cross-border road-sections in a stepwise procedure: we start with including only values for PM10 that come from emission stations within a 750 metre perimeter. We subsequently increase this perimeter by 250 metres for each following regression until a perimeter of two kilometres. The further away from the threshold of one kilometer, we cannot expect the value of PM10 being measured precisely. Our empirical specification thus addresses the trade-off between obtaining a sufficiently large sample for a reliable estimation of our coefficient of interest on the one hand, and measuring our outcome variable PM10 reliably on the other hand. The first factor moreover limits our ability to conduct two-way clustering of standard errors, as our individual road-sections reduce to less than fifty individual clusters. We thus use one-way clustered standard errors at the year-month level.

Figure 6, Panel A depicts our regression result visually for cross-border road-sections (cf. also Table D.1 in the appendix). We observe that for small perimeters, the limited number of observations yields a negative and (partially) significant coefficient until a perimeter of 1.5 kilometers, above which the coefficient becomes insignificantly positive. This is in line with our concern that measurement precision of PM10 decreases increasingly for perimeters larger than one kilometer. Concerning the interpretation, our findings imply that a decrease in the environmental tax differential relative to a neighboring country by one percentage point increases local PM10 emissions at the respective cross-border road-section by between 0.09 and 0.16 $\mu\text{g}/\text{m}^3$. This effect is significant at the 95%-level.

As a placebo test, we also estimate the identical regression for a subset of road-sections located further inside Germany, i.e. at a larger distance from the border. To this end, we choose a set of road orders between 10 and 40, because the road-sections below 10 provide a risk that the associated emission sensor stations are identical to the cross-border road sections. Figure 6, Panel B depicts our placebo regression result visually. There is no discernible effect in any direction of cross-border tax differentials on local PM10 emissions. We thus conclude that environmental tax differentials lead to higher local pollution due to

divergence in truck routes, in line with our findings from the prior section.

5.2 Predicting Emissions with Survey Data

The findings from our second analysis, utilizing the Europe-wide survey data, enable us to quantify the environmental externalities arising from trucks deviating to minimize fuel taxes. We start by considering the excess ton-kilometers driven, as estimated in Table 7. Expanding on our previous findings, we employ a linear prediction model to estimate the cumulative surplus ton-kilometers resulting from tax-induced adjustments within the cross-border trucking sector in Europe. We use the 2SLS point estimate from Table 7 Panel B, along with a lower and upper bound derived from the 95%-confidence interval. We predict the deviation excess-ton kilometers using our sample of journeys throughout Europe. In the subsequent step, we adjust the result based on the sampling rate of the survey data. Finally, we calculate the predicted excess ton-kilometers by multiplying them by the associated tons of CO₂ emissions per ton-kilometer, considering varying levels of CO₂ emission intensity.

The results are displayed in Table 9, showing a range of excess emissions from 426,000 tons to 1,070,000 tons CO₂ emissions caused by tax-induced deviations in truck transport. These additional emissions result from trucks driving longer routes to save on fuel, highlighting an unintended consequence of non-harmonized environmental tax rates on diesel fuel. To contextualize these findings, it's notable that the total yearly CO₂ emissions caused by road transport in the EU amount to 199 million tons (European Environment Agency, 2024). Thus, tax-induced deviation contribute approximately 0.54% of these emissions. Overall, the differences in environmental tax rates on diesel thus lead to an increase in emissions.

6 Discussion and Conclusion

This study quantifies the impact of environmental taxes on commercial truck traffic and its environmental consequences. We use administrative toll data to analyze cross-border truck flows at German borders and assess the response of truck traffic to changes in environmental taxes. Second, we examine individual truck trips across Europe using administrative survey data to investigate whether trucks take detours to avoid high environmental taxes. In both analyses, we use an IV approach to isolate the effect of environmental taxes on the gross price of diesel. Specifically, we instrument the gross price of diesel with the environmental tax to disentangle the changes in the price of diesel induced by environmental taxes. In the second part, we assess the environmental consequences of these tax-induced responses by incorporating data on air pollution and CO₂ emissions.

We find that trucking companies respond significantly to environmental taxes on diesel. While we initially observe a decrease in cross-border truck traffic, especially on heavily congested highways near borders, we do not witness an overall reduction in traffic. Instead, trucking companies drive detours to refuel in countries with lower fuel taxes, leading to increased mileage and exacerbating environmental externalities. Moreover, our investigation into the environmental consequences reveals spikes in air pollution in response to tax-induced changes in commercial traffic at borders. We quantify that the increased truck mileage lead to additional emissions of up to 1 million tons of CO₂.

Our comprehensive investigation into the effects of environmental taxes on commercial truck traffic makes a unique contribution to the existing literature. Responding to the timely and highly relevant call for more research on the impacts of environmental taxes (Lester and Olbert, [2024](#)), our study enhances the understanding of how such taxes influence commercial traffic and provides valuable insights into potential policy solutions to mitigate environmental consequences. The evidence we present suggests that harmonizing tax rates across the EU could be a viable strategy to prevent commercial trucks from deviating to other countries. Such cross-border harmonization could potentially decrease

traffic and foster more sustainable practices in the transportation sector.

Our findings extend beyond Europe and the specific environmental taxes on fuel. Similar variations in fuel taxes exist across US states (Davis and Kilian, [2011](#); Li et al., [2014](#)), and the broader implications of our results can be generalized to scenarios with non-harmonized tax rates on different mobile tax bases. The key takeaway from our study is that when the tax base is mobile and opportunities for tax avoidance exist, there is a risk of shifting behavior. This issue is particularly critical in the context of environmental regulations aimed at reducing emissions to mitigate climate change, as it can lead to leakage and an overall increase in emissions.

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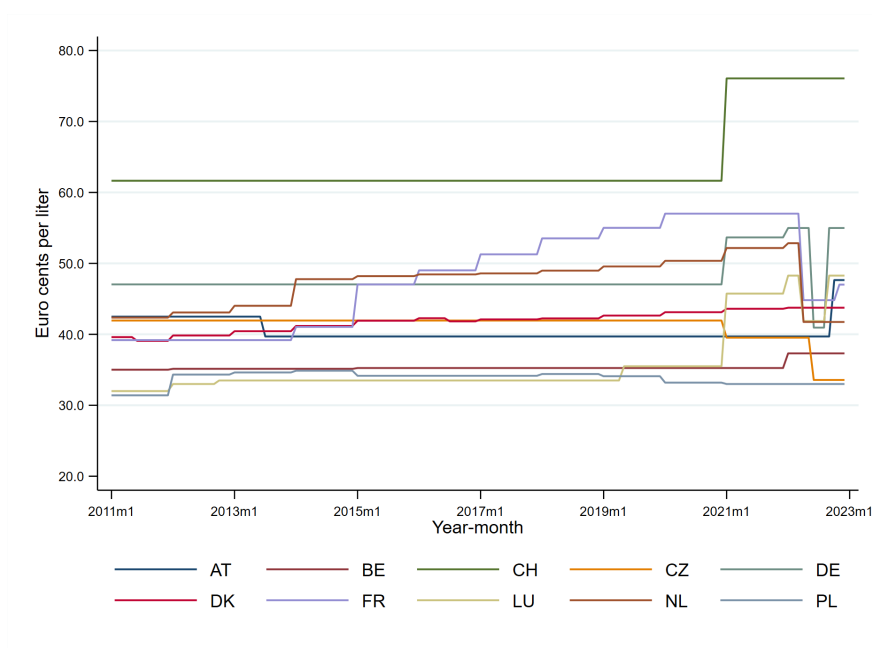
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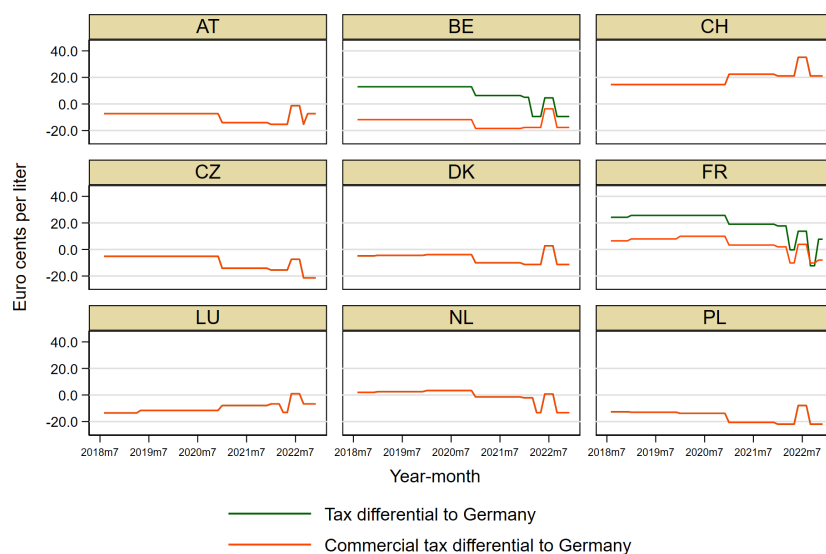
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Figures

Figure 1: Development of environmental taxes on diesel



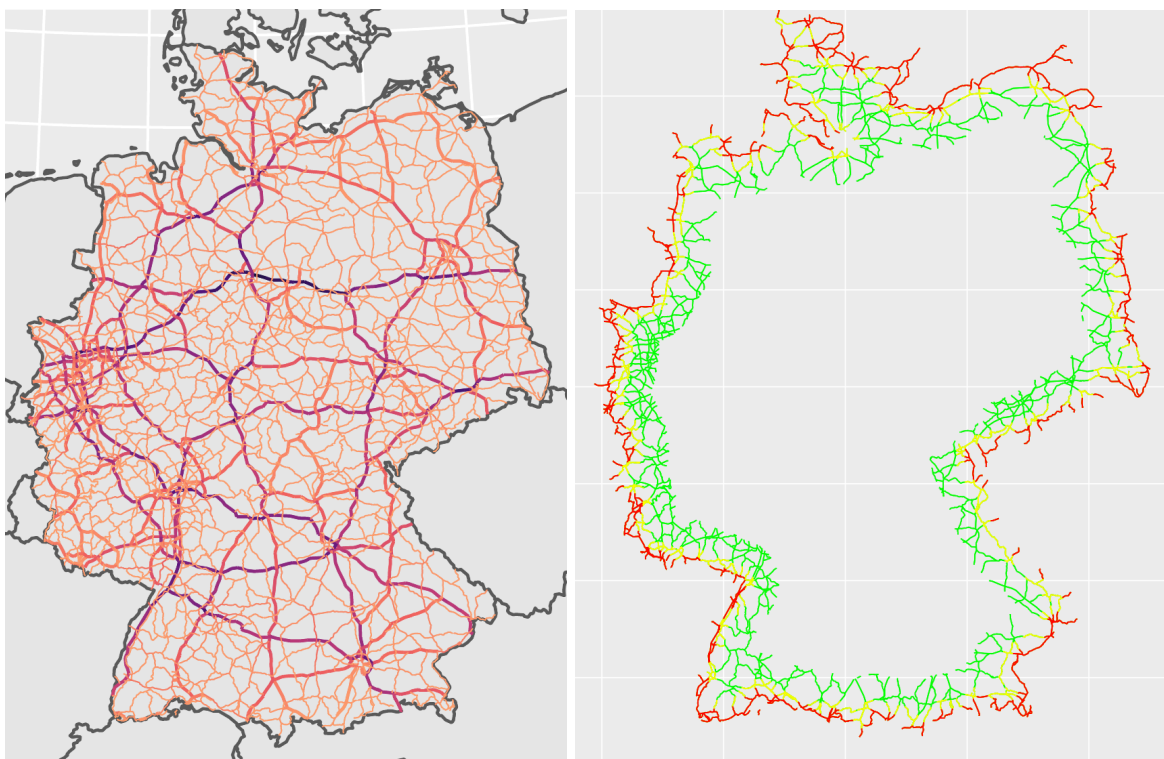
(a) In Germany and neighboring countries



(b) Neighboring countries relative to Germany

Notes: The figures plot the development of environmental taxes on diesel in Germany and its neighboring countries. Panel (a) plots the countries' monthly tax rates in euro cent per liter over the period 2011 to 2022. Panel (b) plots the countries' tax differentials relative to Germany (tax in neighboring country - tax in Germany) in euro cent per liter of diesel fuel over the period mid-2018 to 2022. For Belgium and France that apply different tax rates on commercial vs. private use the green line is the tax differential excluding commercial tax rebates and the orange line is the tax differential including commercial tax rebates.

Figure 2: Geographic plots of road-sections covered by toll data

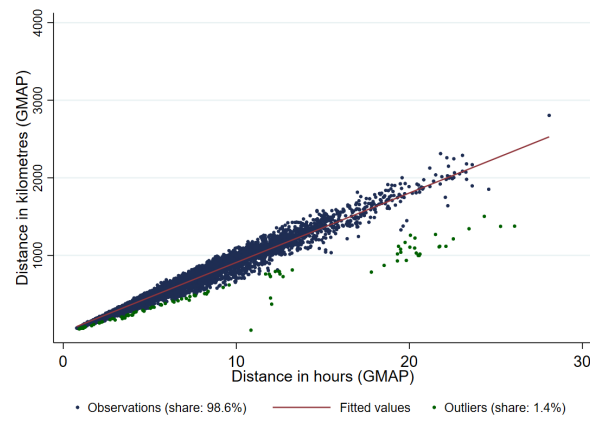


(a) Highways and federal roads in Germany

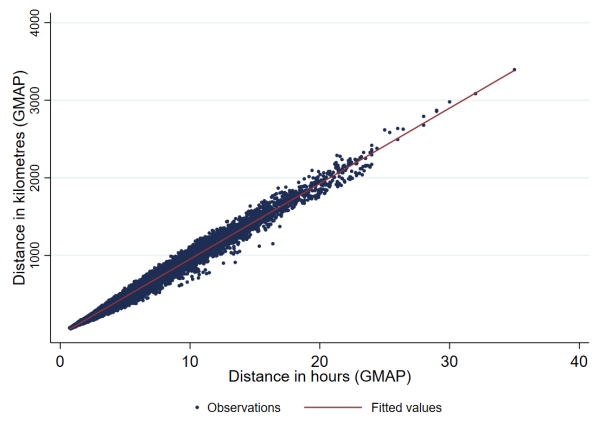
(b) Highways and federal roads at borders

Notes: The figures plot the toll data geographically. Panel (a) shows all highway and federal road road-sections present in the dataset. The thick lines represent highways, and the thin lines represent federal highways. The darker the color, the more traffic a road-section has on average. Panel (b) shows only highways and federal roads around borders. Red lines are all road-sections within 5 kilometers of the border, yellow are all between 5 and 10 kilometers, and green are within 10 to 15 kilometers of the border.

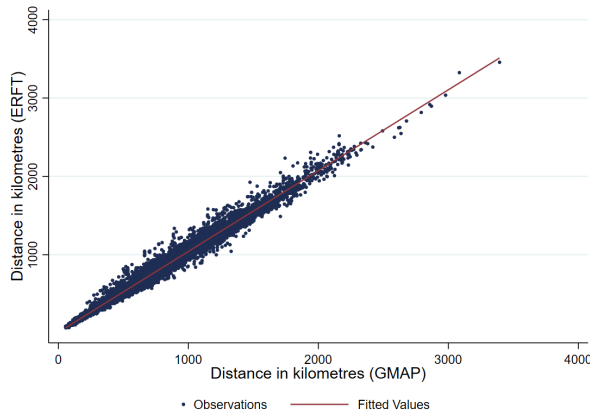
Figure 3: Visualization of GMAP and ERFT data



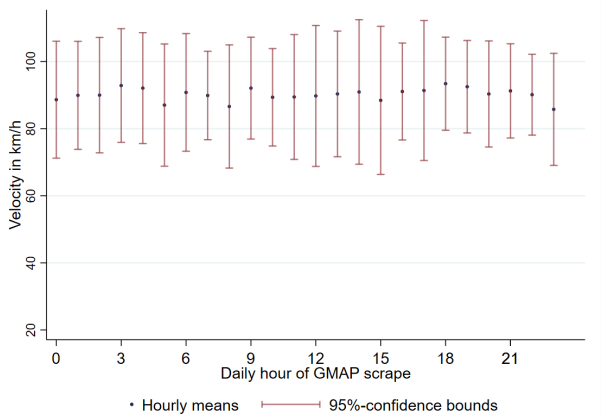
(a) GMAP data before cleaning



(b) GMAP data after cleaning



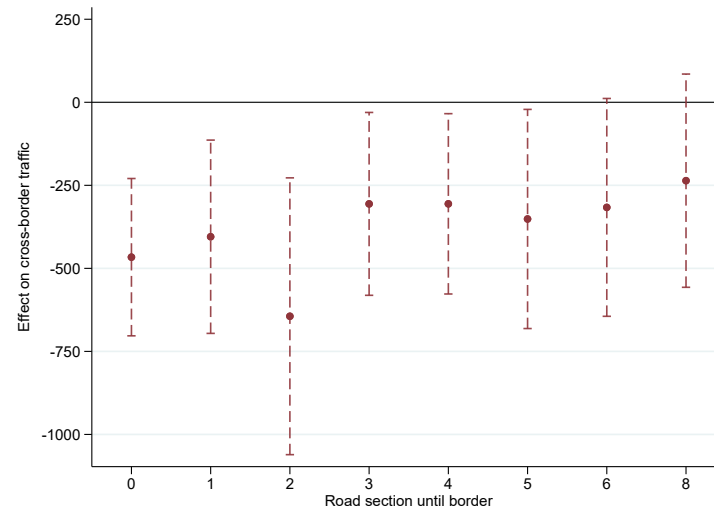
(c) Comparison distance ERFT/GMAP data



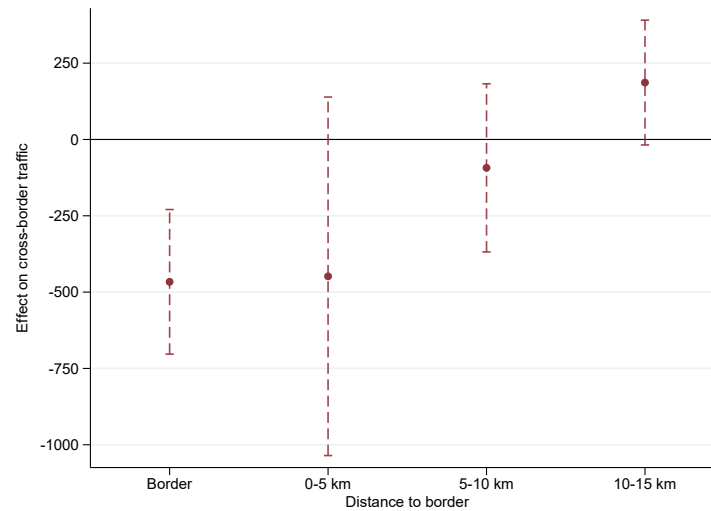
(d) GMAP Daily route velocity

Notes: The figures show properties of the actual ERFT data and the scraped GMAP data. Panels (a) and (b) depict the scraped GMAPs data before and after cleaning. Plotted observations are aggregated at the NUTS2-pair level for the start and end points of each route, provided suchs pairs contain at least 10 observations per route (approx. 95 percent of all routes). Panel (c) provides a comparison of distance lengths between ERFT and GMAP, again aggregated at the level of the NUTS2-start-end-pairs. Panel (d) visualizes the influence of daily scrape hours on average route velocity.

Figure 4: Toll data: Effect by distance to border



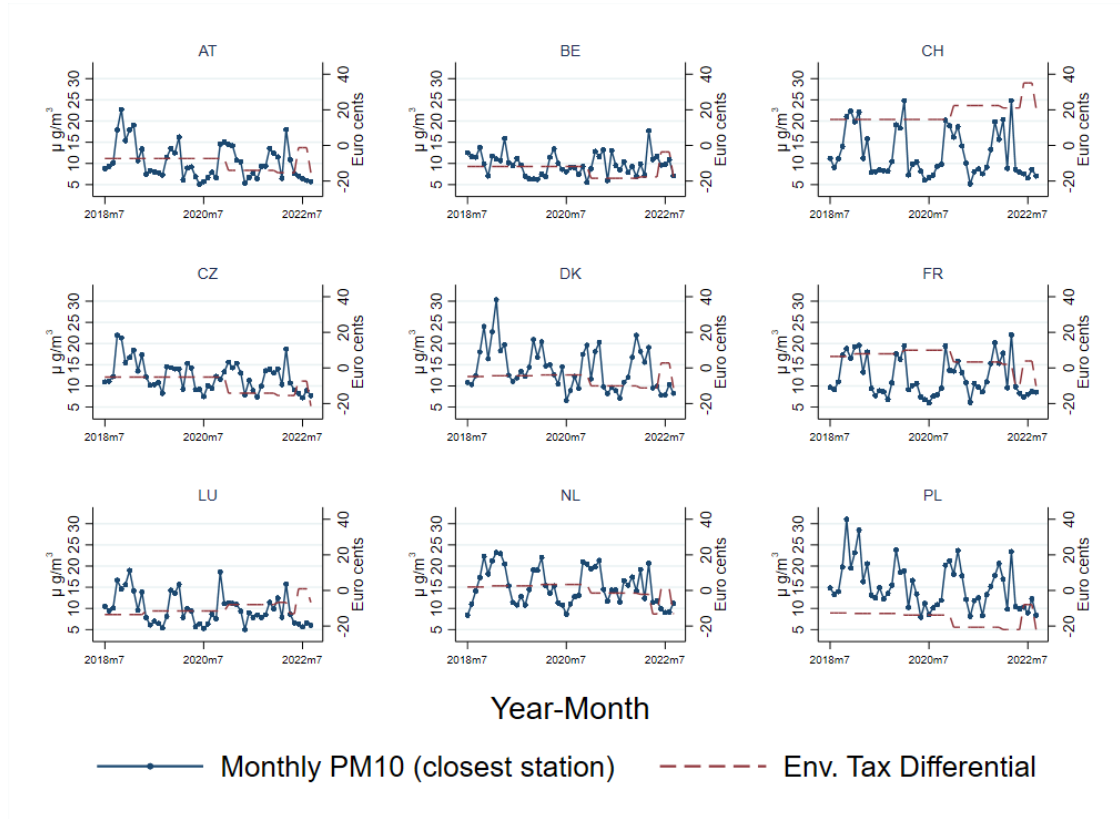
(a) By road-section until border



(b) By distance to border

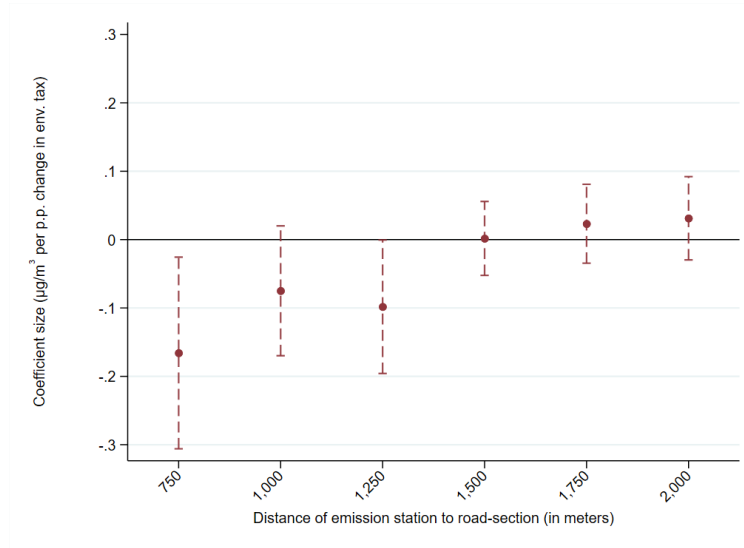
Notes: The figure plots the effect of environmental taxes on commercial highway traffic by road-section and distance to the border. The points plotted are the 2SLS estimates regressing the monthly truck count on the instrumented gross price differential. The gross price differential is calculated relative to Germany (price in neighboring country - price in Germany) and measured in euro cent per liter of diesel fuel. Panel (a) presents the effects by road-section until border. Panel (b) presents the estimates by grouping road-section by their distance to the border. 95% confidence intervals are represented by the dashed red lines and based on standard errors clustered on the road-section level.

Figure 5: Emission data: Environmental tax differentials and emissions

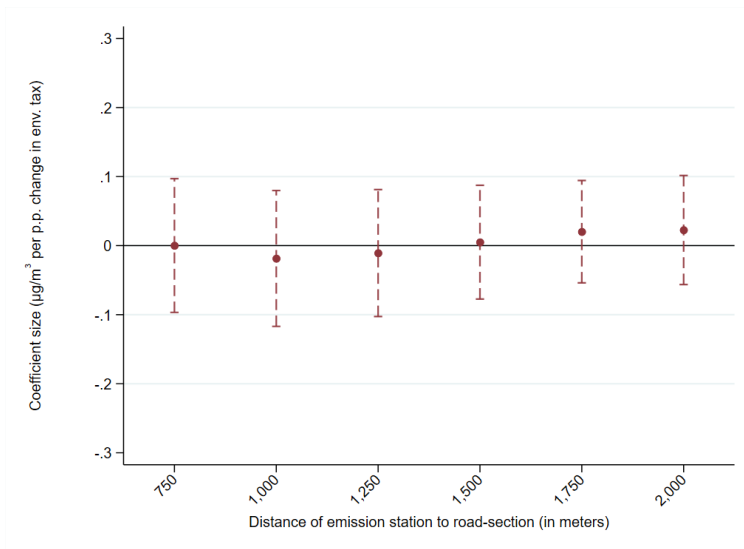


Notes: The figure plots average environmental emissions on cross-border roads and the corresponding environmental tax differential between Germany and its neighboring countries over the time period from July 2018 until September 2022. Monthly PM10 is measured in $\mu\text{g}/\text{m}^3$ for the closest emission station. The environmental tax differential is calculated relative to Germany (tax in neighboring country - tax in Germany) and measured in Euro cents per liter of diesel fuel.

Figure 6: Emission data: Effect of tax differentials on emissions



(a) On cross-border road-sections



(b) On inland road-sections

Notes: The figure plots the effect of environmental taxes on emissions by distance of emission station to road-section. The points plotted are coefficient estimates of a reduced form regression of local PM10 on the environmental tax differential between Germany and its neighboring countries. Panel (a) presents the reduced form estimates for stepwise increases in perimeter size around the cross-border road-sections. Panel (b) repeats the same analysis with a focus on inland road-sections, i.e. those with a road order between 10 and 40. 95% confidence intervals are represented by the dashed red lines and based on standard errors clustered at the year-month level.

Tables

Table 1: Summary statistics of environmental taxes on diesel by country

	Obs.	Mean	St. Dev.	Min.	Max.	Median
Austria (AT)	144	40.45	1.55	39.70	47.65	39.70
Belgium (BE)	144	35.38	0.59	35.02	37.32	35.25
Bulgaria (BG)	144	32.75	0.49	31.44	33.01	33.01
Switzerland (CH)	144	64.05	5.39	61.65	76.07	61.65
Czech Republic (CZ)	144	41.26	1.91	33.57	41.95	41.95
Germany (DE)	144	47.96	2.75	40.95	54.99	47.04
Denmark (DK)	144	41.85	1.37	39.08	43.75	42.11
Estonia (EE)	144	41.54	4.82	37.20	49.30	39.29
Spain (ES)	144	31.29	1.28	30.39	35.19	30.39
Finland (FI)	144	47.18	3.90	36.28	51.25	46.28
France (FR)	144	48.06	6.87	39.19	57.01	49.02
Greece (GR)	144	38.20	3.86	33.00	41.20	41.00
Croatia (HR)	114	39.47	2.18	34.03	40.67	40.67
Hungary (HU)	144	34.36	3.50	22.05	38.88	35.54
Italy (IT)	144	40.13	0.72	37.51	40.32	40.32
Lithuania (LT)	96	35.01	1.83	33.02	37.20	34.70
Luxemburg (LU)	144	35.65	4.67	32.00	48.28	33.50
Latvia (LV)	108	37.04	3.39	33.29	41.40	37.20
Netherlands (NL)	144	47.34	3.29	41.75	52.85	48.45
Poland (PL)	144	33.78	0.93	31.40	34.87	34.16
Portugal (PT)	144	42.22	7.21	27.84	51.33	45.59
Romania (RO)	144	34.67	2.63	30.44	38.45	33.79
Sweden (SE)	144	47.27	7.87	30.29	58.66	46.22
Slovenia (SI)	144	33.00	0.00	33.00	33.00	33.00
Slovakia (SK)	144	37.89	0.91	36.80	38.64	38.64
Total	3486	40.43	8.00	22.05	76.07	39.29

Notes: The table reports summary statistics of (commercial) diesel tax rates (in euro cent per liter) by country during the period 2011 to 2022. The statistics include the number of observations, mean, standard deviation, minimum, maximum and median diesel tax rates by country.

Table 2: Summary statistics of toll data

	Obs	Mean	SD	P1	P50	P99
Road-section length						
Overall	691,098	731	1,272	100	300	6,800
Highway	25,478	5,050	3,493	300	4,300	16,400
Federal road	665,620	565	687	100	300	3,300
Truck count						
Overall	691,098	7,599	17,841	49	2,320	108,671
Highway	25,478	79,273	43,794	4,881	75,079	171,241
Federal road	665,620	4,855	7,273	48	2,182	38,018
Cross-border truck count						
Overall	13,094	17,728	32,222	2	2,786	145,806
Highway	2,550	69,578	40,841	7,898	68,524	161,426
Federal road	10,544	5,188	8,865	2	1,653	46,952

Notes: The table reports summary statistics for the toll data. Displayed are monthly averages spanning the time period mid-2018 till the end of 2022. Road-section length is measured in meters. Truck count is measured in the number of trucks passing the road-section each month. Cross-border truck count restricts the sample to road-sections crossing the border. The number of observations is presented in road-section-months.

Table 3: Summary statistics of survey data

	Obs	Mean	SD	P1	P50	P99
Actual distance (ERFT)						
Overall	1,185,176	624.05	468.91	87	491	2,106
Deviating	71,793	511.22	488.22	69	286	2,108
Optimal distance (GMAP)						
Overall	1,185,176	448.24	368.87	72	323	1,770
Deviating	71,793	473.10	457.95	66	272	1,946
Deviation distance						
Overall	1,185,176	24.39	86.27	-143	10	341
Deviating	71,793	38.12	102.63	-167	22	432
Hired journey						
Overall	1,185,176	0.88	0.32	0	1	1
Deviating	71,793	0.85	0.36	0	1	1
Single-trip						
Overall	1,185,176	0.71	0.45	0	1	1
Deviating	71,793	0.64	0.48	0	1	1
Laden journey						
Overall	1,185,176	0.81	0.39	0	1	1
Deviating	71,793	0.80	0.40	0	1	1
Laden weight						
Overall	1,185,176	136.76	102.33	0	150	313
Deviating	71,793	124.77	103.79	0	114	352

Notes: The table reports summary statistics for the survey data. Displayed are journey averages spanning the time period 2011 till the end of 2020. Overall refers to all cross-border journeys in the dataset, with deviating being those that deviate from the optimal route by crossing another or a different country. Distances are measured in kilometers. The actual distance per journey refers to the ERFT survey data. Optimal distance per journey refers to the fastest route according to GMAP. Deviation is the measured deviation between actual and optimal journey in kilometers. Hired, laden and Single-trip journeys are displayed in percentage points. Laden weight is displayed in 100 kg and corresponds to the weight of the laden goods.

Table 4: Summary statistics of emission data

	Obs	Mean	SD	P1	P50	P99
Distance to emission station						
<i>Full sample</i>						
Overall	401,435	1,554	2	23	547	9,839
Highway	25,209	1,962	1	640	1,452	6,154
Federal road	376,226	1,526	2	23	429	9,839
<i>Cross-border road-sections</i>						
Overall	10,226	8,915	7	251	6,375	28,050
Highway	2,550	7,631	6	668	5,062	26,017
Federal road	7,676	9,341	8	232	6,593	28,481
Monthly emissions (PM10)						
<i>Full sample</i>						
Overall	339,798	12.505	6.6	3.277	11.085	33.127
Highway	21,377	13.221	6.7	3.539	11.928	32.653
Federal road	318,421	12.457	6.6	3.276	11.044	33.138
<i>Cross-border road-sections</i>						
Overall	8,865	12.530	6.8	3.216	11.074	33.180
Highway	2,190	12.873	7.0	3.256	11.116	33.726
Federal road	6,675	12.417	6.7	3.201	11.051	32.653

Notes: The table reports summary statistics for the emission data for the full sample and the cross-border road-sections by road type. The upper half of the table displays distances between the road-sections and the emission sensor stations. The lower half of the table depicts the measured values for PM10. Observations span the time period from mid-2018 until September 2022. Distance between road-sections and emission sensor stations is measured in meters. Monthly PM10 is measured in $\mu\text{g}/\text{m}^3$. The number of observations is presented in road-section-months.

Table 5: Toll data: Effect of environmental taxes on commercial traffic

	Panel A: First stage		Panel B: Reduced form	
	Diesel gross price differential		Cross-border traffic	
	<i>Full sample</i> (1)	<i>Highway</i> (2)	<i>Full sample</i> (1)	<i>Highway</i> (2)
Env. tax differential	0.932*** (0.080)	0.591*** (0.141)	-46.737** (18.091)	-327.138*** (87.593)
<i>Year-month FE</i>	X	X	X	X
<i>Road-section FE</i>	X	X	X	X
<i>Observations</i>	13,094	2,550	13,094	2,550
<i>F-statistic</i>	135.66	17.58	-	-
	Panel C: OLS		Panel D: Second stage	
	Cross-border traffic		Cross-border traffic	
	<i>Full sample</i> (1)	<i>Highway</i> (2)	<i>Full sample</i> (1)	<i>Highway</i> (2)
Gross price differential	7.134 (11.804)	-181.164*** (49.050)	-53.930** (21.828)	-466.317*** (117.849)
<i>Year-month FE</i>	X	X	X	X
<i>Road-section FE</i>	X	X	X	X
<i>Observations</i>	13,094	2,550	13,094	2,550

Notes: The table presents the main results of our toll data analysis. Panel A reports first stage results regressing the environmental tax differential on the gross price differential. Panel B reports estimates of the reduced form equation. The dependent variable is the monthly truck count on a cross-border road-section. The independent variable is the diesel tax differential between Germany and its neighboring countries. Panel C reports results from a OLS specification and Panel D results from the second stage. The dependent variable is the monthly truck count on a cross-border road-section. The independent variable is the (instrumented) gross price differential. The environmental tax differential and the gross price differential are calculated relative to Germany (tax/price in neighboring country - tax/price in Germany) and measured in euro cent per liter of diesel fuel. In all specifications we include road-section and year-month fixed effects. Columns (1) display the results for the full sample, Columns (2) for the highways. Standard errors are clustered on the road-section level and shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Survey data: Effect of environmental taxes on commercial traffic

	Panel A: First stage		Panel B: Reduced form	
	Average diesel gross price		Deviation	
	(1)	(2)	(1)	(2)
Env. tax average	0.43870*** (0.04465)	1.40913*** (0.05370)	0.00429*** (0.00086)	0.00177*** (0.00048)
<i>Year-quarter FE</i>		X		X
<i>NUTS2-start-end FE</i>		X		X
<i>Observations</i>	1,182,316	1,177,246	1,182,339	1,177,269
<i>F-statistic</i>	123.17	649.84	-	-
	Panel C: OLS		Panel D: Second stage	
	Deviation		Deviation	
	(1)	(2)	(1)	(2)
Gross price average	0.00019 (0.00025)	0.00041*** (0.00013)	0.00974*** (0.00286)	0.00126** (0.00039)
<i>Year-quarter FE</i>		X		X
<i>NUTS2-start-end FE</i>		X		X
<i>Observations</i>	1,185,153	1,180,083	1,182,316	1,177,246

Notes: The table presents the results of our survey data analysis estimating a linear probability model. Panel A reports first stage results regressing the average diesel gross price on the average environmental tax on diesel. Panel B reports estimates of the reduced form equation. The dependent variable is the deviation indicator and the independent variable is the average environmental tax on diesel on the journey. Panel C reports results from a OLS specification and Panel D results from the second stage. The dependent variable is the deviation indicator and the independent variable is the (instrumented) average diesel gross price on the journey. Columns (2) include year-quarter and NUTS2-start-end FEs. Robust standard errors are included in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Survey data: Deviation distance

Panel A: Deviation distance			
Excess kilometers			
	<i>2SLS</i> (1)	<i>2SLS</i> (2)	<i>RF</i> (3)
Env. tax average			0.169 (0.196)
Env. tax average x Deviation			0.398*** (0.031)
Gross price average	-2.148*** (0.532)	0.140 (0.137)	
Gross price average x Deviation	0.128*** (0.024)	0.132*** (0.010)	
<i>Year-quarter FE</i>		X	X
<i>NUTS2-start-end FE</i>		X	X
<i>Observations</i>	1,182,316	1,177,246	1,177,269
Panel B: Deviation ton-kilometers			
Excess ton-kilometers			
	<i>2SLS</i> (1)	<i>2SLS</i> (2)	<i>RF</i> (3)
Env. tax average			5.078 (3.052)
Env. tax average x Deviation			6.155*** (0.540)
Gross price average	-37.845*** (9.437)	3.916* (2.106)	
Gross price average x Deviation	1.594*** (0.416)	2.041*** (0.180)	
<i>Year-quarter FE</i>		X	X
<i>NUTS2-start-end FE</i>		X	X
<i>Observations</i>	1,182,316	1,177,246	1,177,269

Notes: The table presents results from our survey dataset estimating the effect of environmental taxes on deviation distance. Panel A reports the results using the deviation distance in kilometers as outcome variable. Panel B reports estimates using deviation in ton-kilometers as outcome variable. The independent variables are the average environmental tax on diesel on the journey and its interaction with the deviation indicator for the reduced form in Columns (3) and the (instrumented) diesel gross price on the journey and its interaction with the deviation indicator for the 2SLS approach in Columns (1) and (2). In Columns (2) and (3) we include year-quarter and NUTS2-start-end FEs. Robust standard errors are included in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Survey data: Country effects

	No. deviations	Log(No. deviations)	Share deviations
	<i>2SLS</i>	<i>2SLS</i>	<i>2SLS</i>
	(1)	(2)	(3)
Gross price average	-2.179 (1.732)	-0.028 (0.022)	-0.007 (0.006)
<i>Year-quarter FE</i>	X	X	X
<i>Country FE</i>	X	X	X
<i>Observations</i>	920	757	920

Notes: The table presents the effects of environmental taxes on deviations by country. The dependent variable is the number of deviations in Column (1), the log(number of deviations) in Column (2) and the share of deviating journeys per country and year-quarter in Column (4). The independent variable is the (instrumented) average diesel gross price per country and year-quarter. All specifications include year-quarter and country FEs. Standard errors clustered on country level are included in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Survey data: Excess emissions from deviations

	Low CO2-Intensity	Medium CO2-Intensity	High CO2-Intensity
Lower Bound	426,540.7	785,732.8	1,046,147.1
Point estimate	431,679.1	795,198.4	1,058,749.8
Upper Bound	436,817.5	804,663.9	1,071,352.5

Notes: The table presents the additional emissions resulting from tax-induced differences in diesel prices across the EU. The estimates are derived from the point estimates and 95%-confidence interval obtained from Table 7, Panel B, using a linear prediction model. The values are then scaled up by the sampling rate and multiplied by the excess ton-kilometers, along with their corresponding tons of CO₂ emissions. We include a low, medium and high CO₂ emissions multiplier.

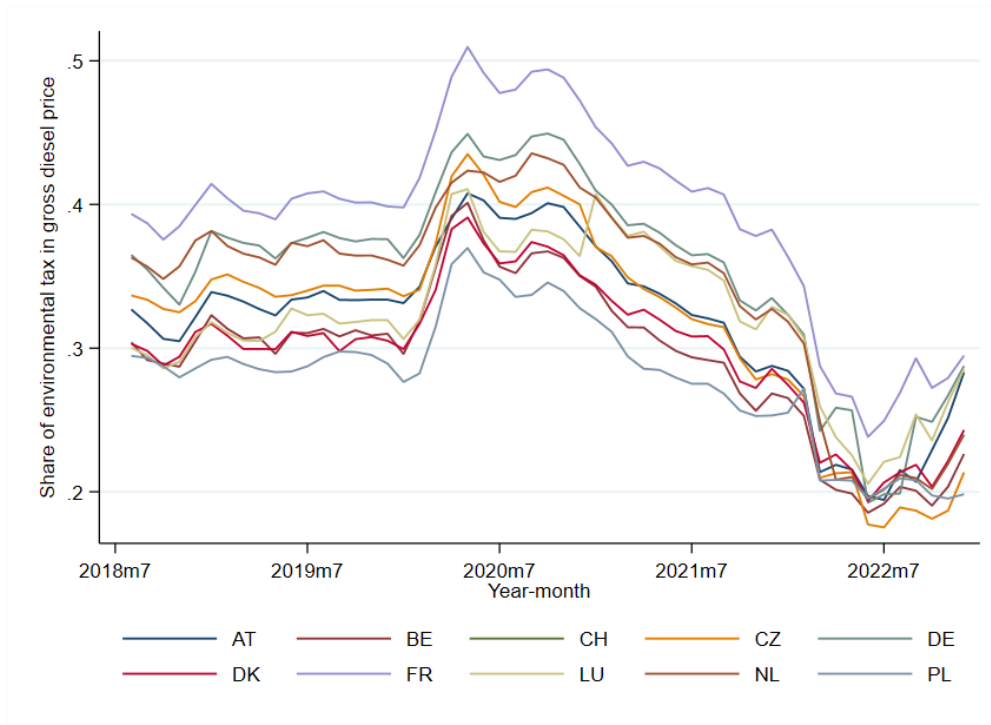
Appendices

The next pages include the following appendices:

- Environmental Taxes and Diesel Price (Appendix [A](#))
- Toll Data (Appendix [B](#))
- Survey Data (Appendix [C](#))
- Emission Data (Appendix [D](#))
- Variable Definitions (Appendix [E](#)).

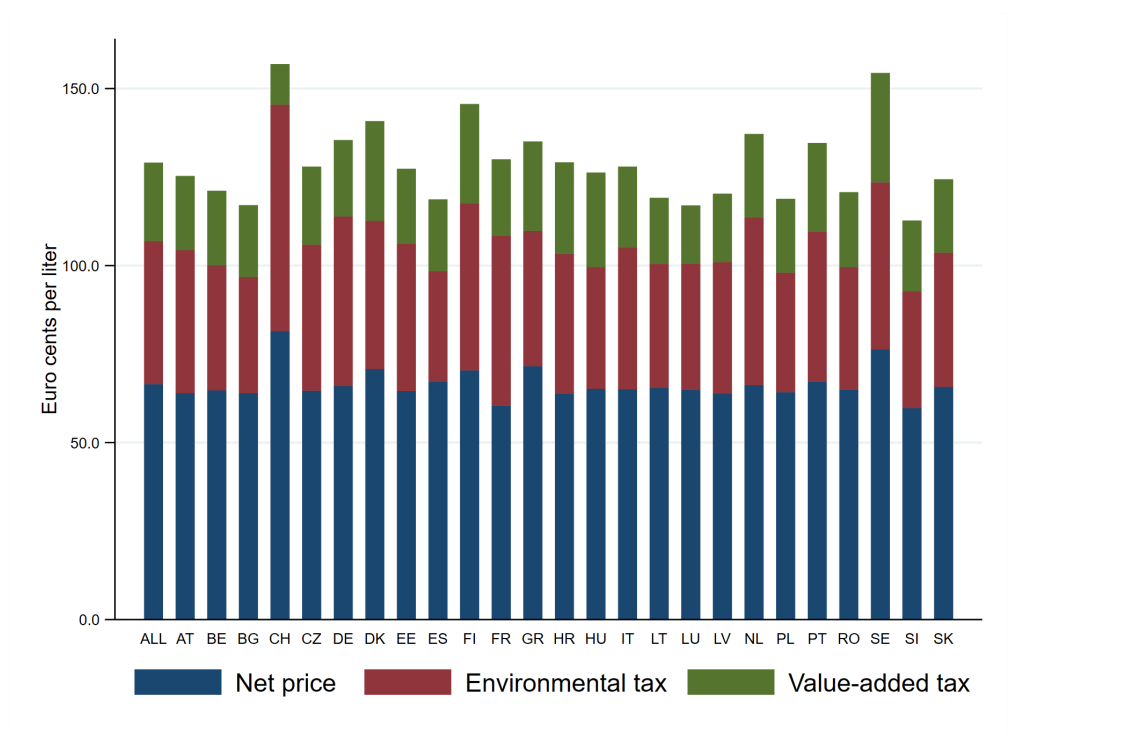
A Environmental Taxes and Diesel Price

Figure A.1: Share of environmental taxes in gross diesel price



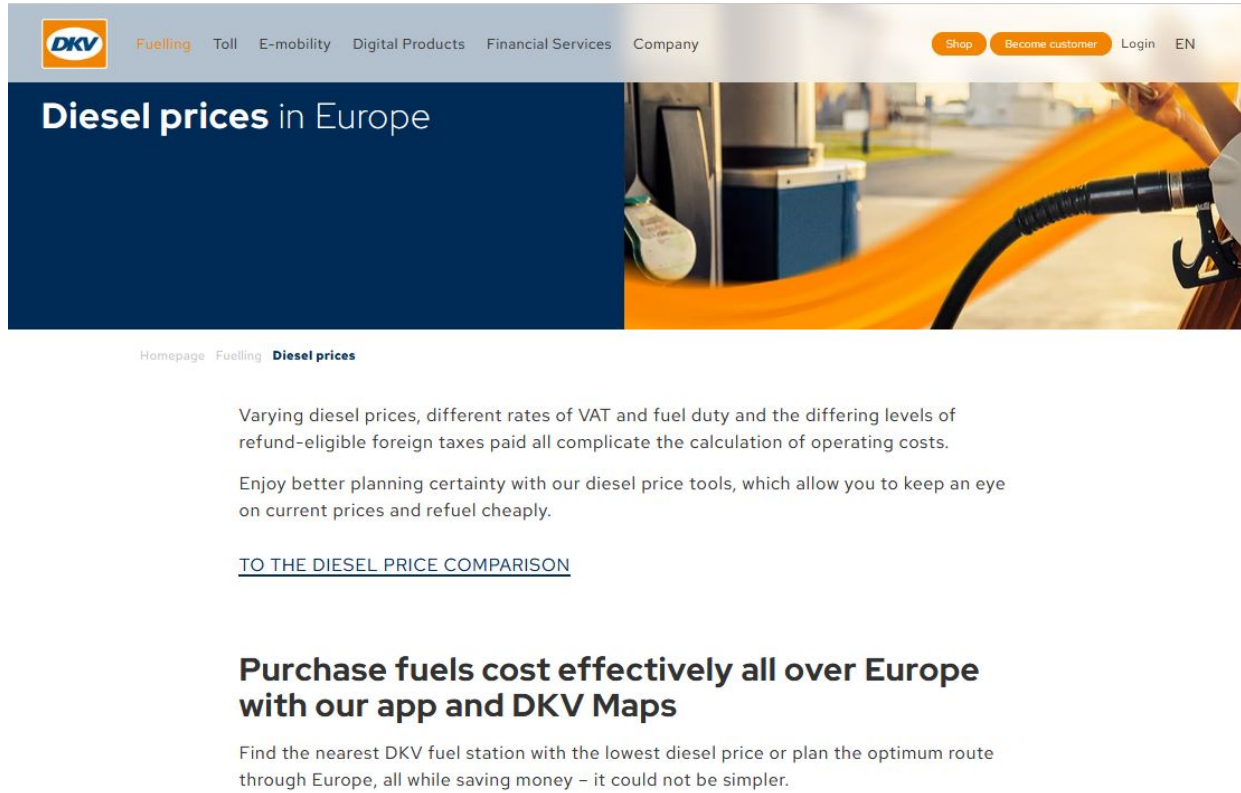
Notes: The figure illustrates the share of environmental taxes in the total gross price of diesel in Germany and its neighboring countries during the period spanning from mid-2018 to 2022.

Figure A.2: Diesel price components by country



Notes: The figure illustrates the components of the total gross price of diesel by continental European countries. The numbers displayed are averages over the time period 2011 to 2022. The components are the net price of diesel, the environmental tax on diesel and the value-added tax. The price components are displayed in euro cents per liter of diesel.

Figure A.3: Advertising diesel price tools for trucking companies



DKV Fuelling Toll E-mobility Digital Products Financial Services Company Shop Become customer Login EN

Diesel prices in Europe

Homepage Fuelling **Diesel prices**

Varying diesel prices, different rates of VAT and fuel duty and the differing levels of refund-eligible foreign taxes paid all complicate the calculation of operating costs.

Enjoy better planning certainty with our diesel price tools, which allow you to keep an eye on current prices and refuel cheaply.

[TO THE DIESEL PRICE COMPARISON](#)

Purchase fuels cost effectively all over Europe with our app and DKV Maps

Find the nearest DKV fuel station with the lowest diesel price or plan the optimum route through Europe, all while saving money – it could not be simpler.

Notes: The figure shows anecdotal evidence how the company DKV Mobility advertises diesel price tools for trucking companies.

Figure A.4: Net and gross diesel price by country



(a) Net price diesel

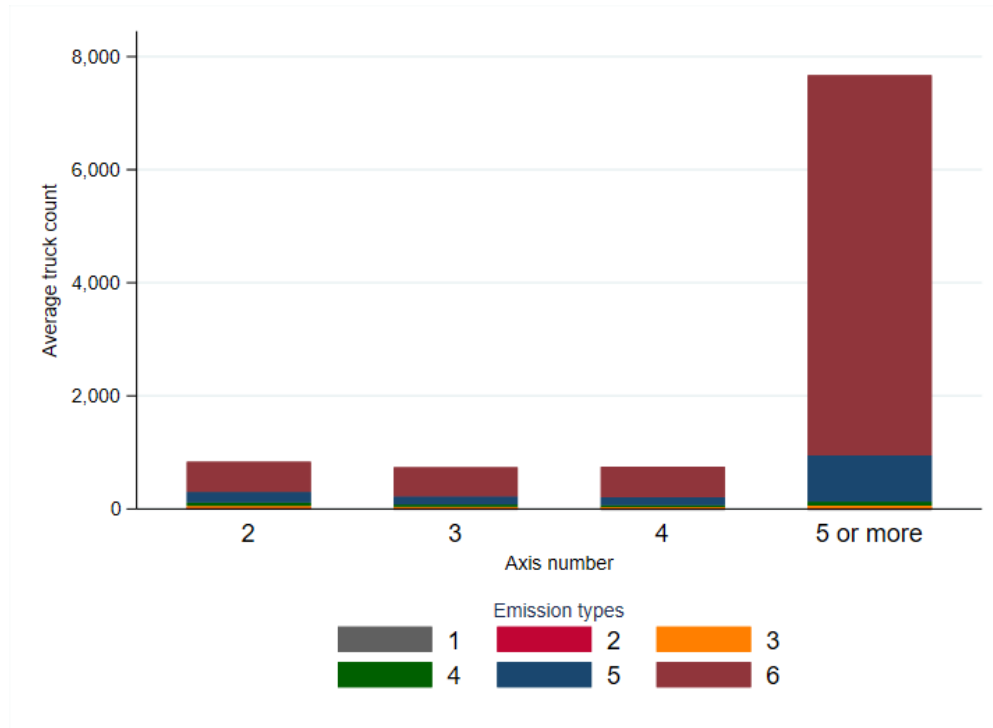


(b) Gross price diesel

Notes: The figure plots the average monthly diesel price in Germany and its neighboring countries for our sample period 2011 to 2022. Panel (a) shows the average monthly net price by country. Panel (b) displays the average monthly gross price including VAT and environmental taxes by country. Prices are displayed in euro cents per liter of diesel.

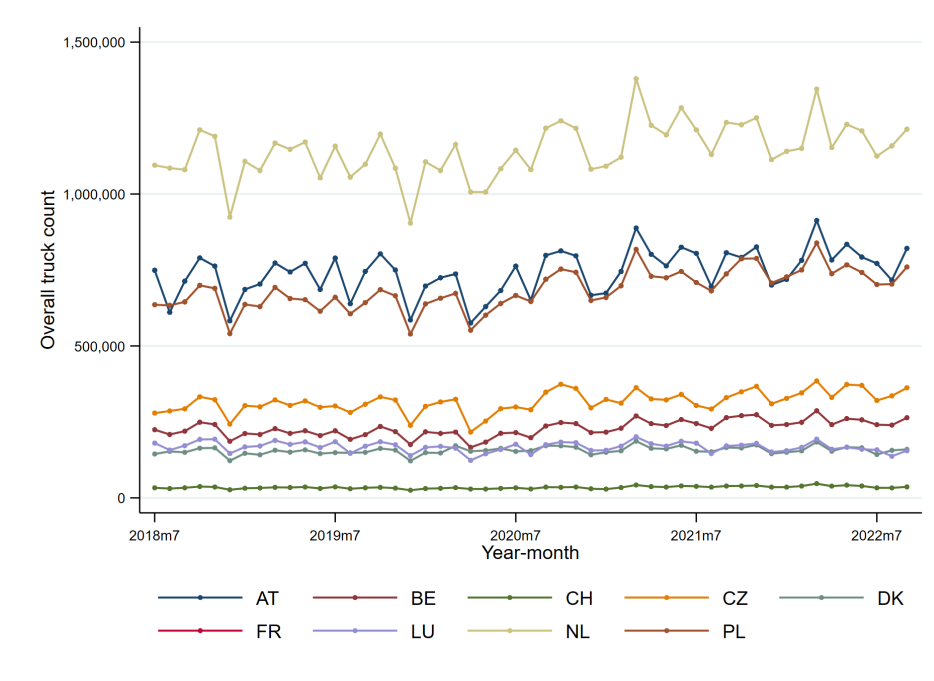
B Toll Data

Figure B.1: Toll data: Average truck count by axes and emission type

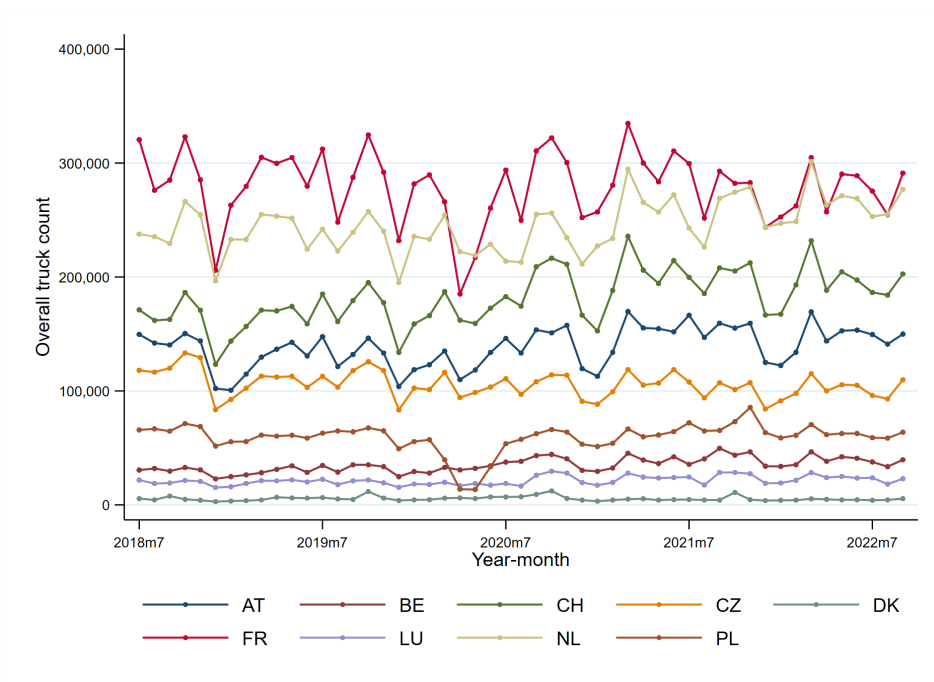


Notes: The figure illustrates the average monthly truck count in our toll data categorized by axis number and emission type. Vehicles with two axes are for example tractors or combined harvesters. Vehicles with three axes are for example car cranes. Trucks in our sample are considered vehicles with more than four axes that are heavily used for road freight transportation. The emission type is based on the european standard for exhaust emissions, Euro 1 to Euro 6. The standards were implemented over time, with Euro 6 being the most recent requirement for newly registered vehicles.

Figure B.2: Toll data: Truck count at border



(a) Highway



(b) Federal road

Notes: The figure plots the monthly overall truck count at the borders to Germany's neighboring countries. Panel (a) illustrates the monthly truck count at cross-border road-section on highways. Panel (b) displays the monthly truck count at cross-border road-section on federal roads.

Table B.1: Toll data: Average monthly truck count by distance to border and country

		Full Sample	Border	Below 5 km	5-10 km	10-15 km
Overall	<i>Mean</i>	7598.59	17727.79	6174.56	6199.36	6469.74
	<i>SD</i>	(17840.57)	(32221.78)	(16257.18)	(13889.69)	(15924.99)
	<i>Obs</i>	691,098	13,094	86,137	72,248	67,178
Austria (AT)	<i>Mean</i>	6031.66	19210.92	7125.17	5649.30	4558.28
	<i>SD</i>	(15317.13)	(35621.10)	(19603.19)	(11698.00)	(10663.45)
	<i>Obs</i>	144,389	2,340	13,846	15,897	17,663
Belgium (BE)	<i>Mean</i>	3663.81	10165.92	3561.25	1807.36	2825.70
	<i>SD</i>	(11303.86)	(24675.20)	(14278.20)	(2595.45)	(8642.24)
	<i>Obs</i>	65,091	1,326	9,984	7,929	7,242
Switzerland (CH)	<i>Mean</i>	6054.93	6765.68	7436.03	6568.55	5141.10
	<i>StD</i>	(6160.11)	(7476.39)	(7616.15)	(6548.51)	(4835.80)
	<i>Obs</i>	77,530	1,632	20,444	14,908	40,546
Czech Republic (CZ)	<i>Mean</i>	6542.46	19319.05	5632.14	5446.97	5883.29
	<i>SD</i>	(13095.61)	(29455.54)	(11010.86)	(13949.86)	(14674.59)
	<i>Obs</i>	89,293	1,116	8,964	7,603	9,781
Denmark (DK)	<i>Mean</i>	7347.47	26952.02	2344.33	4184.25	5366.37
	<i>SD</i>	(15446.31)	(36443.93)	(1354.03)	(2856.95)	(2612.75)
	<i>Obs</i>	20,880	306	1,836	990	1,632
France (FR)	<i>Mean</i>	5481.52	6989.18	4812.66	4697.90	5082.10
	<i>SD</i>	(10744.58)	(12618.05)	(10025.92)	(9632.76)	(9708.28)
	<i>Obs</i>	102,787	2,039	17,118	12,600	7,905
Luxembourg (LU)	<i>Mean</i>	3933.19	11826.22	1402.98	4193.85	3626.22
	<i>SD</i>	(8430.01)	(18383.80)	(1439.32)	(6019.25)	(3874.70)
	<i>Obs</i>	44,873	816	5,428	4,831	6,287
Netherlands (NL)	<i>Mean</i>	18485.58	33117.67	10999.48	11264.38	20448.70
	<i>SD</i>	(32107.47)	(41153.58)	(27004.83)	(23948.83)	(37228.94)
	<i>Obs</i>	92,305	2,142	12,203	10,196	6,836
Poland (PL)	<i>Mean</i>	9058.41	27600.60	7362.24	9457.13	4795.32
	<i>SD</i>	(23800.40)	(46053.79)	(22162.41)	(23624.03)	(6926.83)
	<i>Obs</i>	53,950	1,377	4,790	3,726	3,804

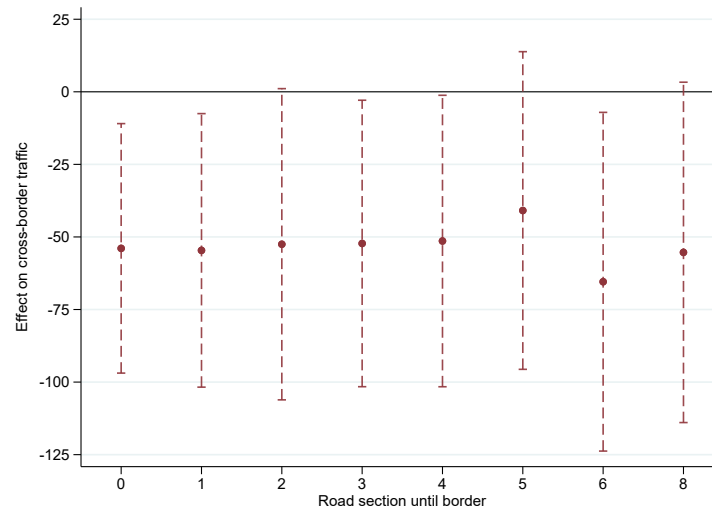
Notes: The table reports average monthly truck counts categorized by road-section and destination country. Full sample includes all road-sections up to 50 km from the border. Border includes only the direct road-section at the border itself. Below 5 km, 5-10 km and 10-15 km comprise all road-sections within the respective corridor distances from the border.

Table B.2: Toll data: Alternative specification

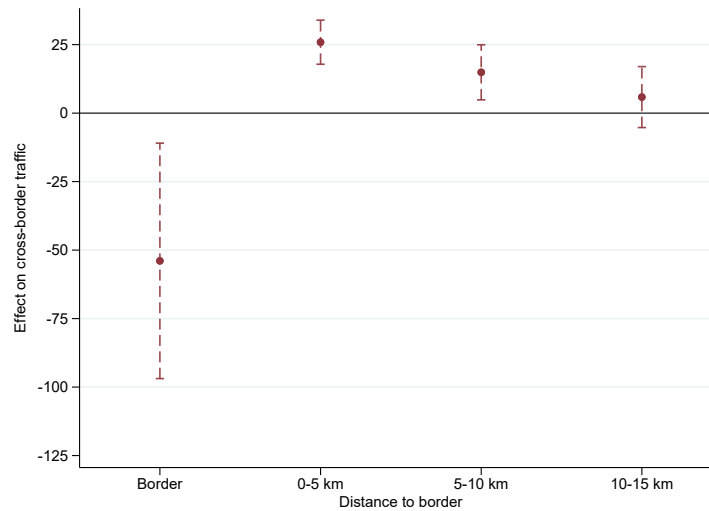
	Log(Cross-border traffic)		
	<i>Highway</i>		
	<i>OLS</i>	<i>2SLS</i>	<i>RF</i>
	(1)	(2)	(3)
Env. tax differential			-0.002 (0.002)
Gross price differential	-0.002*** (0.001)	-0.003*** (0.001)	
<i>Year-month FE</i>	X	X	X
<i>road-section FE</i>	X	X	X
<i>Observations</i>	2,550	2,550	2,550

Notes: The table reports results from the toll data analysis using a log specification for the dependent variable cross-border traffic. Column (1) shows results from an OLS specification and (2) from an IV 2SLS regression. The dependent variable is log of the monthly truck count on a cross-border road-section. The independent variable is the (instrumented) gross price differential. Column (3) shows the results of the reduced form regressing log(Cross-border traffic) on the environmental tax differential. The environmental tax differential and the gross price differential are calculated relative to Germany (tax/price in neighboring country - tax/price in Germany) and measured in euro cent per liter of diesel fuel. In all specifications we include road-section and year-month fixed effects. Standard errors are clustered on the road-section level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure B.3: Toll data: Full sample effect by distance to border



(a) By road-section until border



(b) By distance to border

Notes: The figure plots the effect of environmental taxes on commercial traffic by road-section and distance to the border for the full sample including highways and federal roads. The points plotted are the 2SLS estimates regressing the monthly truck count on the instrumented gross price differential. The gross price differential is calculated relative to Germany (price in neighboring country - price in Germany) and measured in euro cent per liter of diesel fuel. Panel (a) presents the effects by road-section until border. Panel (b) presents the estimates by grouping road-section by their distance to the border. 95% confidence intervals are represented by the dashed red lines and based on standard errors clustered on the road-section level.

Table B.3: Toll data: Robustness Covid-19

	Panel A: Second stage		Panel B: Reduced form	
	Cross-border traffic		Cross-border traffic	
	<i>Full sample</i>	<i>Highway</i>	<i>Full sample</i>	<i>Highway</i>
	(1)	(2)	(1)	(2)
Env. tax differential			-44.254** (18.049)	-326.573*** (88.811)
Gross price differential	-49.237** (20.815)	-459.794*** (111.871)		
<i>Year-month FE</i>	X	X	X	X
<i>road-section FE</i>	X	X	X	X
<i>Observations</i>	12585	2450	12585	2450

Notes: The table reports robustness tests of the toll data analysis excluding all months (032020 - 04020) that are affected by the Covid-19 pandemic. Panel A shows the results from the second stage regression. Column (2) the results from the reduced form. The dependent variable is the monthly truck count on a cross-border road-section. The independent variable is the instrumented gross price differential in Panel A and the environmental tax differential in Panel B. The environmental tax differential and the gross price differential are calculated relative to Germany (tax/price in neighboring country - tax/price in Germany) and measured in euro cent per liter of diesel fuel. Columns (1) show the results for the full sample, Columns (2) for highways. In all specifications we include road-section and year-month fixed effects. Standard errors are clustered on the road-section level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.4: Toll data: Robustness Ukraine war

	Panel A: Second stage		Panel B: Reduced form	
	Cross-border traffic		Cross-border traffic	
	<i>Full sample</i>	<i>Highway</i>	<i>Full sample</i>	<i>Highway</i>
	(1)	(2)	(1)	(2)
Env. tax differential			-68.992*** (24.746)	-484.053*** (99.859)
Gross price differential	-165.044*** (61.591)	-552.531*** (146.755)		
<i>Year-month FE</i>	X	X	X	X
<i>road-section FE</i>	X	X	X	X
<i>Observations</i>	11034	2150	11034	2150

Notes: The table reports robustness tests of the toll data analysis excluding all months (022022 - 12022) that are affected by the Ukraine. Panel A shows the results from the second stage regression. Column (2) the results from the reduced form. The dependent variable is the monthly truck count on a cross-border road-section. The independent variable is the instrumented gross price differential in Panel A and the environmental tax differential in Panel B. The environmental tax differential and the gross price differential are calculated relative to Germany (tax/price in neighboring country - tax/price in Germany) and measured in euro cent per liter of diesel fuel. Columns (1) show the results for the full sample, Columns (2) for highways. In all specifications we include road-section and year-month fixed effects. Standard errors are clustered on the road-section level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.5: Toll data: Highway effect by country

		Cross-border traffic		
		Second stage (1)	Reduced form (2)	Observations (3)
Austria (AT)	<i>Env. tax differential</i>		-708.090*** (145.250)	510
	<i>Gross price differential</i>	-335.246*** (65.176)		510
Belgium (BE)	<i>Env. tax differential</i>		-670.545** (133.209)	204
	<i>Gross price differential</i>	-411.137*** (70.559)		204
Switzerland (CH)	<i>Env. tax differential</i>		157.447 (137.209)	102
	<i>Gross price differential</i>	-206.558 (126.654)		102
Czech Republic (CZ)	<i>Env. tax differential</i>		-882.522*** (60.238)	204
	<i>Gross price differential</i>	-403.141*** (23.772)		204
Denmark (DK)	<i>Env. tax differential</i>		-453.825** (9.138)	102
	<i>Gross price differential</i>	-209.418*** (2.967)		102
Luxembourg (LU)	<i>Env. tax differential</i>		-207.071*** (34.003)	204
	<i>Gross price differential</i>	9285.859*** (1317.286)		204
Netherlands (NL)	<i>Env. tax differential</i>		-514.451*** (89.554)	816
	<i>Gross price differential</i>	-291.295*** (49.067)		816
Poland (PL)	<i>Env. tax differential</i>		-1090.498*** (65.176)	408
	<i>Gross price differential</i>	-509.069*** (205.568)		408

Notes: The table reports results of the toll data analysis by country on highways. Column (1) shows the results from the second stage regression. Column (2) the results form the reduced form. The dependent variable is the monthly truck count on a cross-border road-section. The independent variable is the instrumented gross price differential in the second stage specification and the environmental tax differential in the reduced form specification. The environmental tax differential and the gross price differential are calculated relative to Germany (tax/price in neighboring country - tax/price in Germany) and measured in euro cent per liter of diesel fuel. Standard errors are clustered on the road-section level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Survey Data

Table C.1: Survey data: Meta-information

	Global sampling rate	Response rate (2012)
Austria (AT)	0.8	98.5
Belgium (BE)	0.9	65.3
Bulgaria (BG)	0.2	69.8
Switzerland (CH)	0.3	58.6
Czech Republic (CZ)	0.3	91.9
Germany (DE)	0.4	96.3
Denmark (DK)	0.4	99.0
Estonia (EE)	0.7	75.5
Spain (ES)	0.3	94.3
Finland (FI)	0.1	58.1
France (FR)	0.3	77.9
Greece (GR)	0.1	83.8
Croatia (HR)	0.8	80.1
Hungary (HU)	1.3	87.5
Italy (IT)	0.4	25.5
Lithuania (LT)	0.6	90.1
Luxembourg (LU)	3.3	91.9
Latvia (LV)	0.5	78.4
Netherlands (NL)	0.5	76.6
Norway (NO)	0.3	95.6
Poland (PL)	0.1	84.7
Portugal (PT)	0.9	75.4
Romania (RO)	0.7	97.3
Sweden (SE)	0.4	70.1
Slovenia (SI)	0.7	74.5
Slovakia (SK)	0.1	87.5
Average	0.7	80.5

Notes: The table reports meta-data about the European Road Freight Transport Statistics. The figures are obtained from the 2014 EUROSTAT publication "Methodologies used in surveys of road freight transport in Member States, EFTA and Candidate Countries". The global sampling rate is expressed as a percentage across time (yearly weeks) and space (countries). Response rate is also a percentage.

Table C.2: Survey data: Probit specification

	Panel A: Second stage	Panel B: Reduced form
	Deviation	Deviation
	(1)	(1)
Env. tax average		0.034*** (0.007)
Gross price average	0.054*** (0.007)	
<i>Observations</i>	1,182,316	1,182,339

Notes: The table reports results from the survey data analysis using a probit model. Panel A shows results from the second stage regression, Panel B from the reduced form. The dependent variable is the deviation indicator and the independent variable is the instrumented average diesel gross price on the journey for the second stage and the environmental tax differential for the reduced form. Robust standard errors are included in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

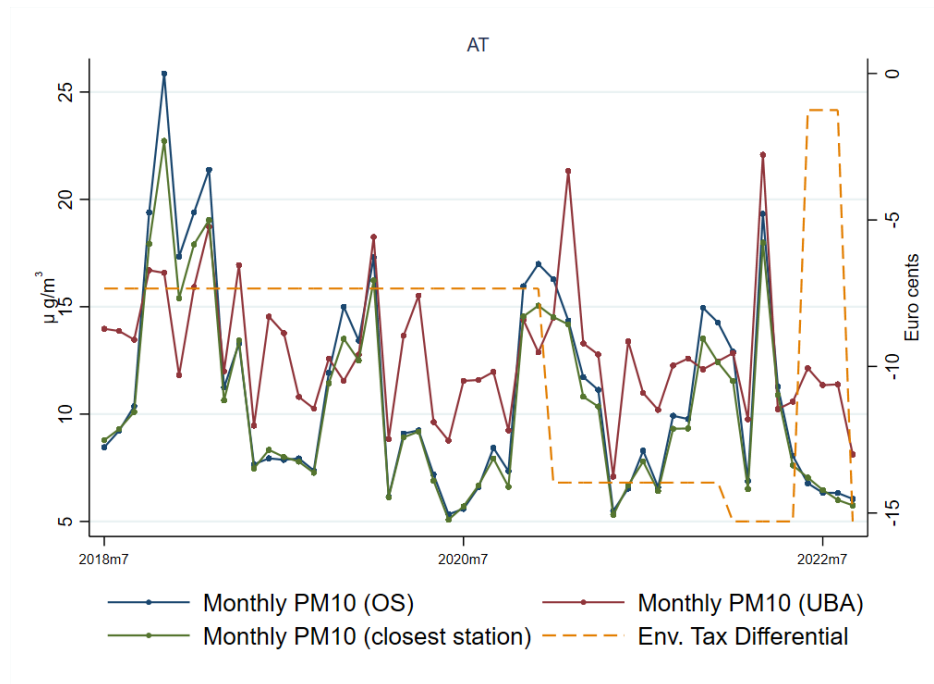
Table C.3: Survey data: Share of routes by country

	Share overall routes	Share deviating routes	Ratio per country
Austria (AT)	0.110	0.279	2.531
Belgium (BE)	0.352	0.414	1.177
Bulgaria (BG)	0.001	0.000	0.000
Switzerland (CH)	0.043	0.075	1.733
Czech Republic (CZ)	0.072	0.092	1.274
Germany (DE)	0.522	0.709	1.359
Denmark (DK)	0.013	0.001	0.095
Estonia (EE)	0.007	0.000	0.052
Spain (ES)	0.056	0.012	0.223
Finland (FI)	0.001	0.000	0.285
France (FR)	0.242	0.223	0.919
Greece (GR)	0.000	0.000	0.000
Croatia (HR)	0.008	0.015	1.887
Hungary (HU)	0.065	0.069	1.064
Italy (IT)	0.074	0.113	1.538
Lithuania (LT)	0.022	0.006	0.295
Luxembourg (LU)	0.075	0.135	1.801
Latvia (LV)	0.019	0.004	0.199
Netherlands (NL)	0.377	0.487	1.291
Norway (NO)	0.000	0.000	0.000
Poland (PL)	0.070	0.062	0.882
Portugal (PT)	0.019	0.002	0.130
Romania (RO)	0.009	0.014	1.522
Sweden (SE)	0.003	0.000	0.171
Slovenia (SI)	0.037	0.045	1.220
Slovakia (SK)	0.047	0.068	1.443
Observations	1,285,633	80,687	1,285,633

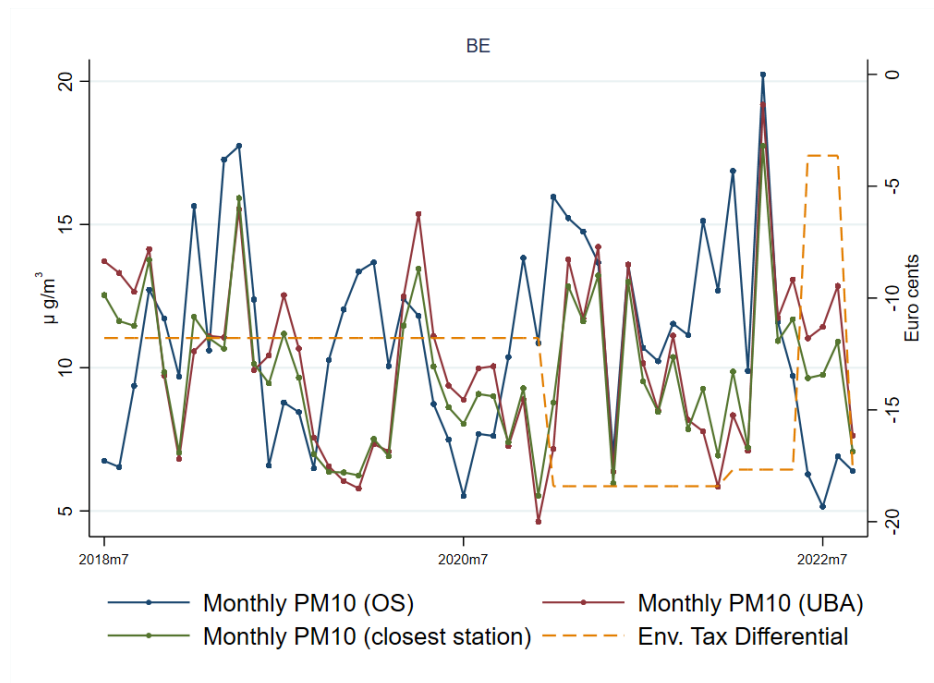
Notes: This table shows the country share of truck traffic routes in Europe. The first column provides the share of overall journeys passing through a particular country. The second column provides the same statistic, but only for the subset of routes which diverge from the optimal route calculated via GMAP. The third and last column contains the ratio between the first two columns, indicating whether a specific country is part of detours relative to its overall importance in international truck traffic.

D Emission Data

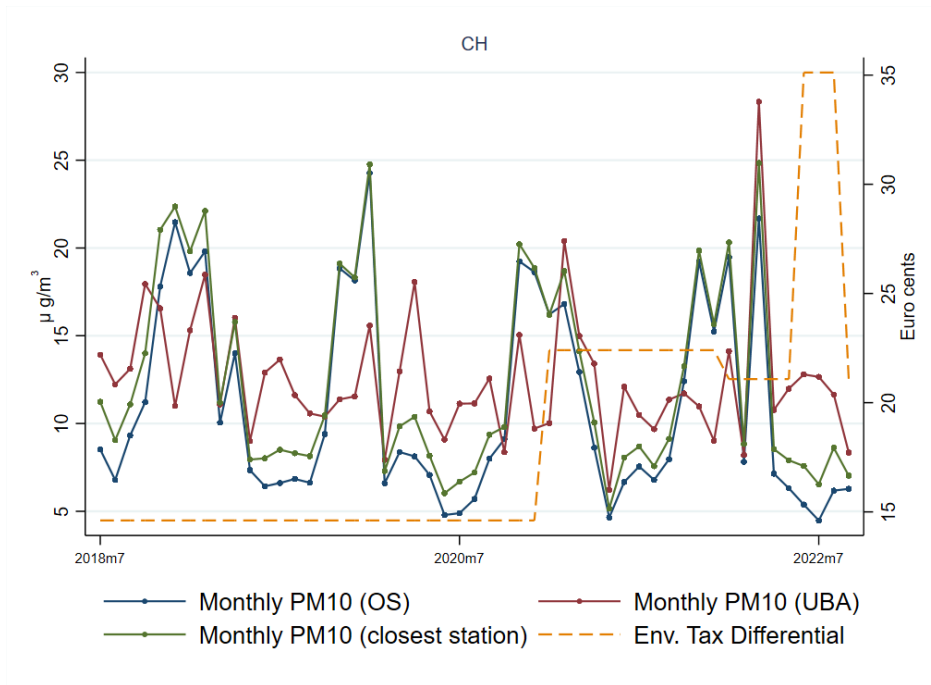
Figure D.1: Emission data: Environmental tax differentials and emissions by border-country



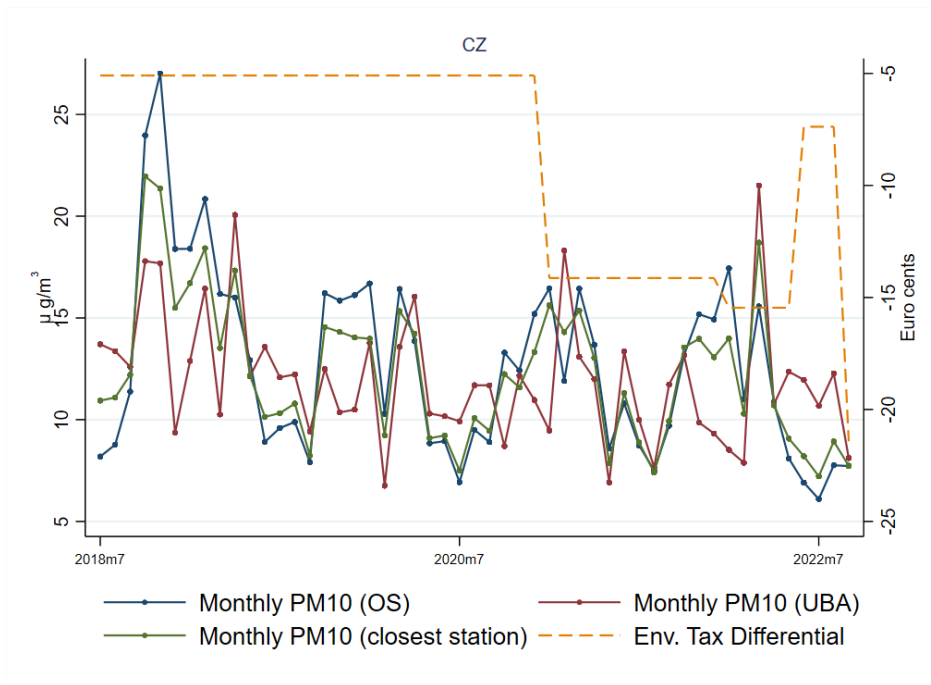
(a) Austria



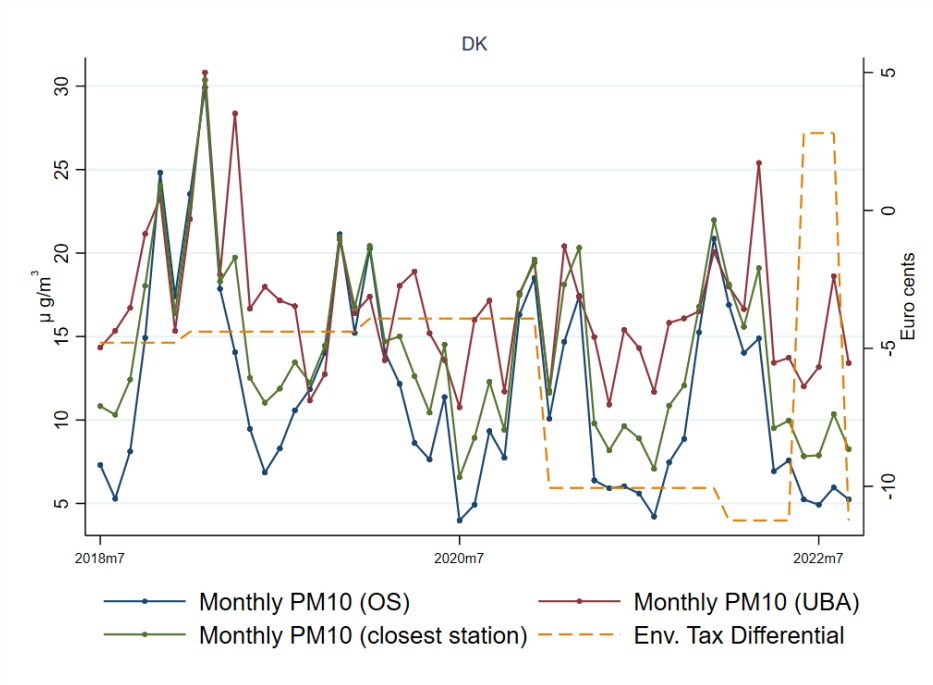
(b) Belgium



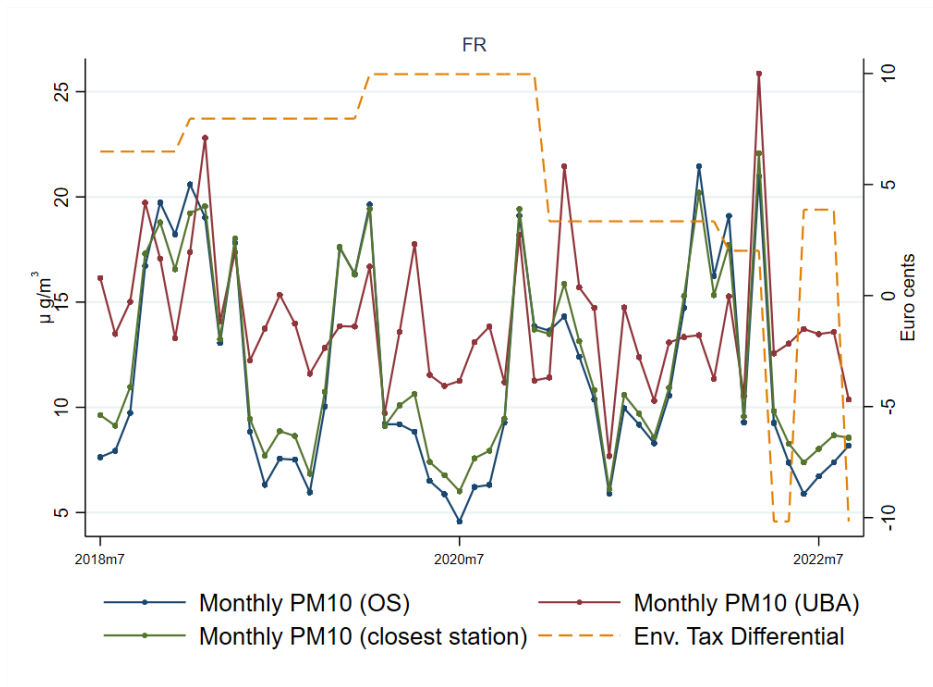
(c) Switzerland



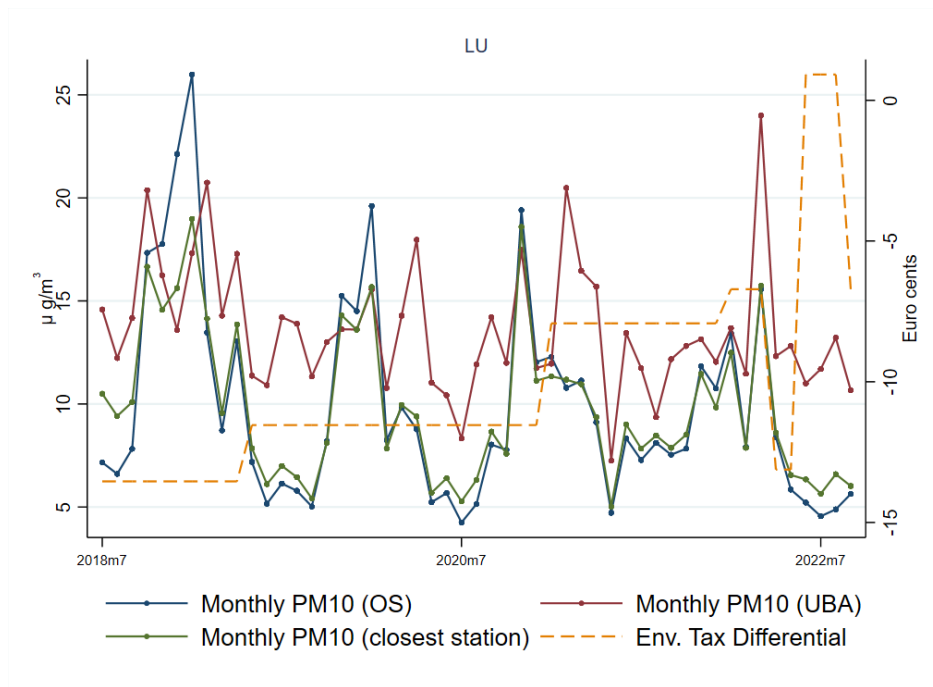
(d) Czech Republic



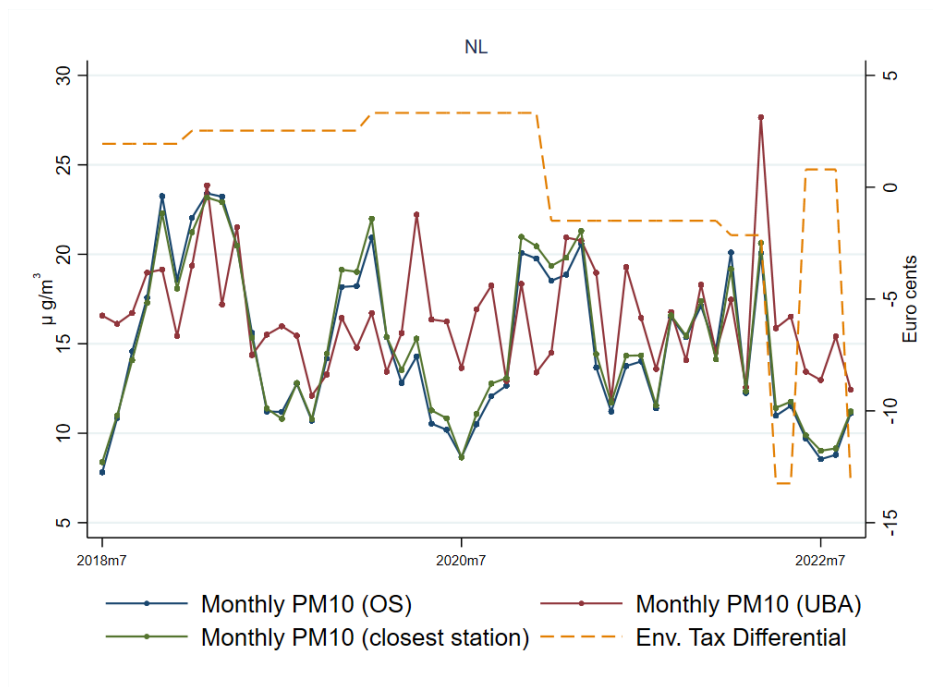
(e) Denmark



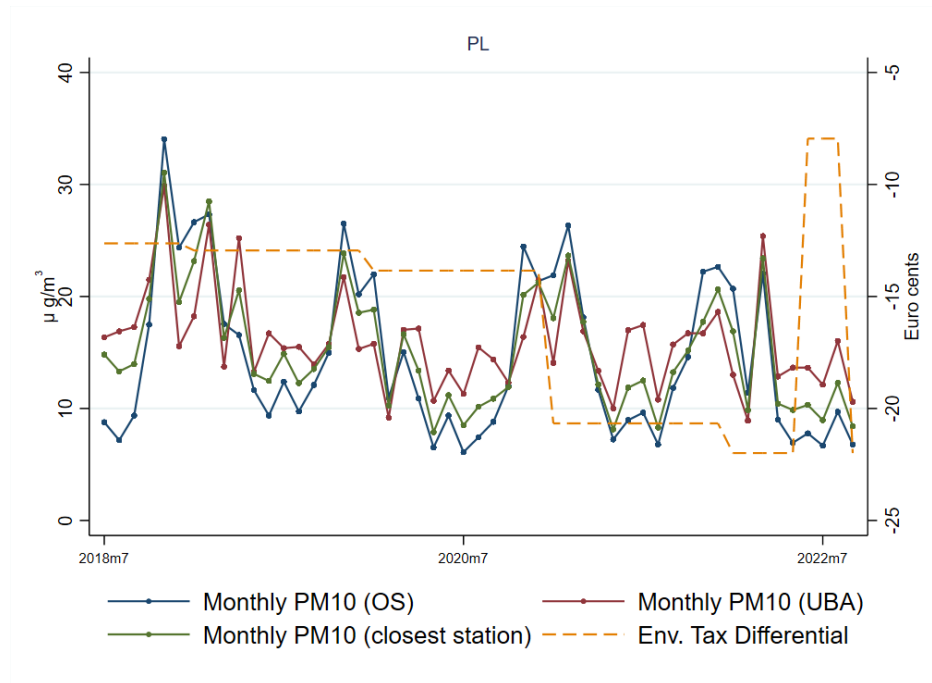
(f) France



(g) Luxembourg



(h) Netherlands



(i) Poland

Notes: The figure plots average environmental emissions on cross-border roads and the corresponding environmental tax differential between Germany and its neighboring countries over the time period from July 2018 until September 2022. Monthly PM10 is measured in $\mu\text{g}/\text{m}^3$, and shown for each data source separately. The environmental tax differential is relative to Germany (tax in neighboring country - tax in Germany) and measured in euro cent per liter of diesel fuel.

Table D.1: Emission data: effect of tax differential on local pollution

	PM10					
Env. tax differential	-0.166** (0.07)	-0.0749 (0.05)	-0.0982** (0.05)	0.00165 (0.03)	0.0231 (0.03)	0.0312 (0.03)
Constant	13.46*** (0.51)	13.02*** (0.15)	12.77*** (0.09)	12.52*** (0.02)	12.96*** (0.05)	13.23*** (0.11)
Meters	750	1000	1250	1500	1750	2000
Adj. R^2	0.62	0.67	0.65	0.70	0.70	0.70
Obs. Number	278	493	539	885	1,168	1,689

Notes: The table reports reduced form estimates of regressing local, cross-border measures of PM10 on cross-border environmental tax differentials using OLS. Meters refers to the road-section distance to the closest emission sensor station. One-way clustered standard errors are reported in parentheses. All columns include road-section and year-month fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Variable Definitions

Variable	Definition
Toll data	
<i>Gross price differential</i>	Difference in the diesel gross price between Germany's neighboring country c and Germany in year-month m .
<i>Env. tax differential</i>	Difference in environmental tax rate on diesel between Germany's neighboring country c and Germany in year-month m .
<i>Cross-border traffic</i>	The number of trucks passing a road-section i in year-month m .
Survey data	
<i>Gross price average</i>	Average diesel gross price on journey j in year-month m .
<i>Env. tax average</i>	Average environmental tax rate on journey j in year-month m .
<i>Deviation</i>	Indicator variable equal to one if the truck deviated on a journey in distance and to another country.
Emission data	
<i>Emissions</i>	PM10 emissions measured by the closest sensor station to road-section i in year-month m .